
ADAPTIVE MULTI-VALUE CONTROL IN LLMs VIA CAUSAL ACTIVATION STEERING

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ABSTRACT

Large language models (LLMs) are increasingly deployed in settings where responses must reflect multiple, potentially interacting social norms and human values. Activation steering offers a lightweight alternative to training-based alignment by modifying internal activations at inference time. However, prior human-value steering methods have largely considered values in isolation, while direct composition of multiple directions relies on fixed intervention strengths that cannot respond to the model’s evolving internal state. Motivated by this key observation, we introduce AIMES, a framework for adaptive multi-value activation steering within a single generation. AIMES constructs layer-specific bipolar directions for moral-foundation values and uses intermediate-layer vocabulary readouts as online observers. An observer-guided controller then adapts the strength of each requested value intervention at every decoding step based on its current observed state, without training a separate value-state estimator. Across multiple instruction-tuned model families, value combinations, and intervention depths, we find that multi-value controllability varies across both value combinations and intervention locations. Compared with fixed joint steering and prompt-based steering, AIMES shows depth-dependent advantages that are broadly supported across two independent evaluators, with some variation in the precise depth at which specific control effects emerge. These advantages come with smaller realized activation-space interventions than fixed-joint steering and comparable response quality. Overall, our results suggest that online observer feedback can provide lightweight, state-aware adaptation for single-pass multi-value steering.

1 INTRODUCTION

Large Language Models (LLMs) are increasingly deployed across critical domains such as education (Al Faraby & Romadhony, 2024; Alhafni et al., 2024), research (Ren et al., 2025; Liao et al., 2024), healthcare (Yang et al., 2023; Cascella et al., 2023), and finance (Lakkaraju et al., 2023; Zhao et al., 2024), where model behavior may need to reflect multiple, potentially interacting human values and social norms. This challenge is closely related to pluralistic alignment, which seeks to accommodate diverse and potentially competing values and perspectives, including through steerable adaptation to specified value trade-offs (Sorensen et al., 2024). Existing alignment methods include reinforcement-learning-based approaches such as RLHF (Ouyang et al., 2022), RLAIIF (Lee et al., 2023), TRPO (Schulman et al., 2015), and PPO (Schulman et al., 2017), as well as reward-model-free methods such as DPO (Rafailov et al., 2023). However, these approaches generally require additional preference collection, optimization, or fine-tuning, making post-hoc adaptation to changing value requirements costly.

Activation steering (Subramani et al., 2022; Turner et al., 2023; Zou et al., 2023; Li et al., 2023) provides a lightweight inference-time alternative mechanism for controlling model behavior by modifying internal representations without updating model parameters. Recent work has extended this idea to human values, showing that moral concepts can be captured by layer-specific activation directions and causally influenced through intervention (Yu et al., 2026), but primarily considers one value at a

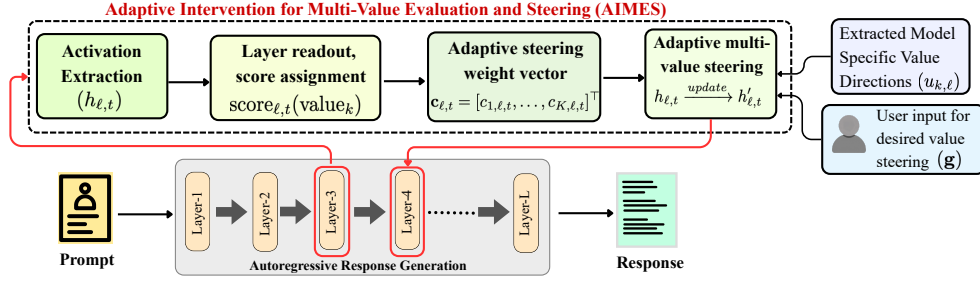


Figure 1: **Overview of AIMES.** At each decoding step, the hidden state activation ($h_{\ell,t}$) at the intervention layer is mapped to value-specific observer scores, which determine adaptive steering weights ($c_{\ell,t}$) for the requested values. These weights modulate precomputed model and layer-specific value directions ($u_{k,\ell}$) before generation continues, enabling closed-loop multi-value steering.

time. Jointly steering multiple values is more challenging because their representations need not be independent: directions may be correlated or conflicting, and intervention along one value can affect the expression of others. Similar interference has been observed in generic multi-attribute activation steering, where composing several directions can reduce control reliability (van der Weij et al., 2024; Nguyen et al., 2025b; Ghasemi et al., 2026; Jiang et al., 2025). Existing approaches to multi-value alignment largely address this problem through other mechanisms, such as prompting (Kim et al., 2026) or combining separately generated value-conditioned outputs (Zheng et al., 2026). As a result, joint activation-level control of multiple human values remains comparatively underexplored. A natural extension to multiple values is to jointly apply several value directions with fixed strengths, but such open-loop steering cannot adapt to changes in the model’s internal state during the generation process. Recent work instead formulates activation steering as feedback control, using estimates of the current concept state to adjust intervention strength (Nguyen et al., 2025a; Bharadwaj, 2025; Prokopiou et al., 2026; Skifstad et al., 2026). Nevertheless, these methods typically rely on separately trained probes, classifiers, or dynamical models. Intermediate-layer vocabulary readouts, including *Logit Lens* (nostalgebraist, 2020), *Tuned Lens* (Belrose et al., 2023), and *J-Lens* (Gurnee et al., 2026), offer a lightweight alternative by exposing information directly from hidden representations. However, representational accessibility does not necessarily imply causal steerability (Billa, 2026; Nadaf, 2026), motivating an empirical comparison of candidate readouts before using them as feedback signals for adaptive control.

Motivated by these observations, we introduce AIMES (Adaptive Intervention for Multi-Value Evaluation and Steering), a framework that turns joint multi-value activation steering into a state-aware feedback process (Figure 1). Rather than composing multiple value directions with fixed strengths, AIMES observes their current expression in an intermediate representation and adjusts their relative strength during decoding. This enables adaptive coordination of interacting value directions without training a separate value-state estimator. Across multiple models, value objectives, and intervention depths, our results show that multi-value controllability depends on both value combination and intervention location, and the cross-judge analysis reveals robust advantages in some depth regions. Our contributions are:

- **Bipolar value directions and online observation.** Building on contrastive activation steering (Rimsky et al., 2024; Yu et al., 2026), we construct model- and layer-specific bipolar human-value directions for bidirectional value control. Intermediate-layer vocabulary readouts are used as online value-state observers without training a separate state estimator.
- **Adaptive multi-value steering.** We formulate joint value steering as a closed-loop control problem and introduce an observer-guided controller that adapts the relative strength of multiple value-specific interventions at every decoding step.
- **Depth-aware and cross-judge empirical evaluation.** Across five models spanning three families and parameter scales, and multiple multi-value objectives, we systematically evaluate adaptive control across intervention depth under two independent judges, GPT-5.6-Sol and Claude Opus-4.8, characterizing the robustness and evaluator sensitivity of the observed multi-value control patterns.

2 RELATED WORK

In this section, we review prior work on activation steering and vocabulary readout methods, and additional background is provided in Appendix A.

Activation Steering for Human Values and Multiple Attributes. Activation steering (Subramani et al., 2022) modifies model behavior at inference time by intervening along directions identified in hidden representation space, without updating model parameters (Turner et al., 2023; Zou et al., 2023; Li et al., 2023). In the human-value setting, (Yu et al., 2026) construct layer-specific contrastive directions for the five dimensions of Moral Foundations Theory (*MFT*) (Graham et al., 2013) and show that interventions along these directions can causally alter foundation-relevant behavior. Importantly, they also observe cross-value effects, suggesting that value directions cannot generally be treated as independent. This issue becomes more pronounced when multiple directions are applied jointly. More broadly, multi-attribute steering methods such as *MAT-Steer* (Nguyen et al., 2025b), *ORBIT* (Ghasemi et al., 2026), *MSRS* (Jiang et al., 2025), and *K-Steering* (Oozeer et al., 2025) address interference among jointly controlled attributes using mechanisms including gating, orthogonalization, subspace decomposition, rotation, and nonlinear direction construction. These methods are motivated in part by evidence that naive vector composition becomes less reliable as the number of controlled attributes increases (van der Weij et al., 2024). However, the resulting intervention is typically determined before generation and does not adapt to the evolving expression of the controlled attributes during decoding.

Steering Multiple Human Values. Recent work has explored multi-value control through prompting, activation-level, and compositional approaches: *VALUEFLOW* (Kim et al., 2026) composes natural-language value specifications and shows that value interactions can be reinforcing or competing, while *VISPA* (Zheng et al., 2026) combines separately generated value-conditioned outputs rather than jointly steering multiple directions within a single generation. The paper (Yu et al., 2026) constructs moral-foundation activation directions but intervene on values individually. Joint activation-level control of multiple human values within a single generation therefore remains comparatively underexplored. Inspired by these approaches, we compare *AIMES* with two non-adaptive alternatives: *Fixed Multi-Value Steering*, which jointly applies multiple value directions with pre-determined strengths, and *prompt steering*, which specifies the same multi-value objective through natural-language instructions without modifying internal activations. Fixed joint steering is closely related to generic multi-attribute activation steering, where vector composition can introduce interference and reduce controllability (van der Weij et al., 2024; Nguyen et al., 2025b; Ghasemi et al., 2026; Jiang et al., 2025). Concurrent work by (Pan et al., 2026) attributes such interference to geometric entanglement and proposes a static Gram-matrix correction. In contrast, *AIMES* uses online observer feedback to adapt value-specific intervention strengths during generation.

Closed-Loop Activation Steering. Recent work formulates activation steering as a feedback-control problem, adapting intervention strength to the model’s evolving internal state. (Nguyen et al., 2025a) distinguish fixed-strength open-loop steering from feedback-based control using the discrepancy between desired and observed concept states; (Bharadwaj, 2025) use PID-style feedback with a chunk-level classifier during reasoning; (Prokopiou et al., 2026) study feedback-controlled multi-attribute steering in symbolic music generation; and (Skifstad et al., 2026) model layer-wise computation as a locally linear dynamical system and derive interventions using linear-quadratic regulation. Relatedly, *ODESteer* (Zhao et al., 2026) makes the steering direction state-dependent through the gradient of a learned density-ratio barrier, but focuses on single-attribute control and requires fitting the barrier. A key requirement in observer-based closed-loop steering is an *observer* that estimates the controlled state, typically through a separately trained classifier, probe, or fitted dynamical model (Nguyen et al., 2025a; Bharadwaj, 2025; Prokopiou et al., 2026; Skifstad et al., 2026). *AIMES* instead uses intermediate-layer vocabulary readouts as training-free online observers and adapts the strengths of multiple jointly applied value directions during generation.

Intermediate-Layer Vocabulary Readouts. To interpret the intermediate states, intermediate-layer readout methods map transformer hidden states into vocabulary-aligned coordinates: the *Logit Lens* (nostalgebraist, 2020) directly applies the output projection, the *Tuned Lens* (Belrose et al., 2023) learns layer-specific transformations toward the final output space, and *J-Lens* (Gurnee et al.,

2026) uses layer-specific Jacobian information to approximate downstream vocabulary-aligned content. Beyond interpretability, (Billa, 2026) show that Logit-Lens-based accessibility can help predict concept steerability and effective intervention depth across models and concept families. However, representational visibility does not necessarily imply causal controllability: effective steering directions can remain weakly exposed by intermediate readouts such as the Logit Lens (Billa, 2026; Nadaf, 2026). This distinction is especially important in closed-loop steering, where the readout itself can potentially serve as the feedback signal for intervention.

3 AIMES: OBSERVER-GUIDED ADAPTIVE MULTI-VALUE STEERING

In this section, we present our main proposed framework, AIMES (Adaptive Intervention for Multi-Value Evaluation and Steering), which consists of an offline value direction-construction stage and an online adaptive-steering stage. Offline, we construct model and layer-specific human-value directions $u_{k,\ell}$ for each target value V_k (Rimsky et al., 2024; Yu et al., 2026). During generation, intermediate-layer observers (Gurnee et al., 2026; nostalgebraist, 2020) estimate the current expression of the controlled values and guide the allocation of value-specific steering strengths. Relative to Fixed Multi-Value Steering, AIMES replaces fixed coefficients with observer-guided adaptive weights that are updated online. The AIMES framework has three major components: (1) *value-specific steering directions*, which define the intervention axes; (2) an *intermediate-layer value observer*, which estimates the current value state from a vocabulary-space readout; and (3) an *observer-guided controller*, which adjusts the contribution of each requested value direction during generation. The intervention layer is fixed within each generation run and varied across experiments to study the effect of intervention depth.

3.1 VALUE-SPECIFIC STEERING DIRECTION ESTIMATION

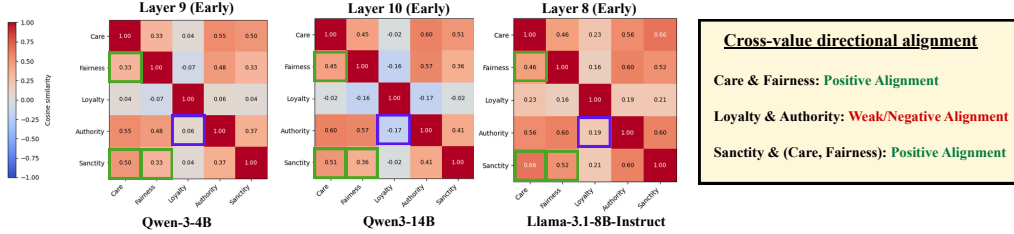


Figure 2: **Cross-value alignment analysis.** The figure shows early-layer cosine similarities for three representative models, highlighting positive Care-Fairness alignment, weaker Loyalty-Authority alignment, and positive Sanctity alignment with Care and Fairness.

Given an autoregressive language model M with L layers, let $\ell \in \{1, \dots, L\}$ denote the layer at which an intervention is applied during a given generation run. For an input prompt x_i , the model generates a response $y_i = (y_i^1, \dots, y_i^{T_i})$, where T_i is the number of generated tokens. We consider a set of K target human values $\{V_1, \dots, V_K\}$ and construct each value-specific steering direction from a matched contrastive corpus, following the mean-difference principle commonly used in activation steering (Rimsky et al., 2024; Yu et al., 2026). For each value V_k , we form paired sets $\mathcal{D}_k^+$ and $\mathcal{D}_k^-$, where each pair describes the same underlying scenario while expressing opposite poles of the target value dimension. For the Moral Foundations Theory dimensions considered here (Graham et al., 2013), these contrasts are *Care/Harm*, *Fairness/Cheating*, *Loyalty/Betrayal*, *Authority/Subversion*, and *Sanctity/Degradation*. The complete vignette-generation, pairing, and validation procedure is described in Appendix C.2. Formally, at layer ℓ , we compute the positive and negative mean activations, $\mu_{k,\ell}^+$ and $\mu_{k,\ell}^-$, respectively, and define the normalized value direction $u_{k,\ell}$ as:

$$\mu_{k,\ell}^+ = \frac{1}{|\mathcal{D}_k^+|} \sum_{x \in \mathcal{D}_k^+} h_\ell(x), \quad \mu_{k,\ell}^- = \frac{1}{|\mathcal{D}_k^-|} \sum_{x \in \mathcal{D}_k^-} h_\ell(x), \quad u_{k,\ell} = \frac{\mu_{k,\ell}^+ - \mu_{k,\ell}^-}{\|\mu_{k,\ell}^+ - \mu_{k,\ell}^-\|_2}. \quad (1)$$

Since the positive and negative sets consist of one-to-one matched pairs of equal size, the mean-difference direction is equivalent to the average pairwise activation difference. The resulting direction $u_{k,\ell}$ therefore defines a bipolar activation-space axis, with its positive and negative orientations corresponding to the two poles of value V_k . Following *Contrastive Activation Addition* (CAA) (Rimsky et al., 2024), this matched construction provides a direct bidirectional intervention axis while reducing variation due to topic, actors, setting, and writing style. This is particularly useful for AIMES, which supports both amplification and suppression along the same value dimension. However, the existence of a separable direction does not imply that its behavior is identical across layers. We therefore evaluate multi-value steering across multiple intervention depths in Section 4.

Geometry-Guided Value-Objective Selection. For the multi-value steering analysis, we select three objectives to represent distinct interaction regimes suggested by the learned value geometry. As illustrated in Figure 2, *Care* and *Fairness* are positively aligned, motivating ($\uparrow$ Care $\uparrow$ Fairness) as an aligned coordination setting. *Loyalty* and *Authority* show substantially weaker and, in some models, negative alignment, motivating the ($\uparrow$ Loyalty $\uparrow$ Authority) objective as a weakly-coupled setting. These two pairings have also been considered in prior multi-value MFT steering (Kim et al., 2026) and are consistent with the broader individualizing-binding distinction in MFT (Graham et al., 2011; 2013). Finally, *Sanctity* is positively aligned with both *Care* and *Fairness*; we therefore consider ($\uparrow$ Care $\uparrow$ Fairness $\downarrow$ Sanctity) as a competing setting, where a value geometrically coupled to the promoted pair is intentionally steered in the opposite direction. Full similarity matrices across five models and depth regions are provided in Appendix C.4.

3.2 INTERMEDIATE-LAYER VALUE OBSERVER: DESIGN AND IMPLEMENTATION

To adapt steering strength during generation, AIMES requires an estimate of how strongly each target value V_k is expressed in the model’s current intermediate state. Rather than training a separate value classifier, we derive this estimate directly from intermediate-layer vocabulary readouts. We consider existing vocabulary readout methods (Gurnee et al., 2026; nostalgebraist, 2020) as candidate observers, and use J-Lens as the default based on the ablation in Appendix C.3. No value-specific estimator is trained: the readout is fixed for a given model and layer and is shared across all values and steering objectives. Logit Lens uses the model’s unembedding matrix W_U , while J-Lens additionally uses a precomputed layer-specific Jacobian J_ℓ ; further details are provided in Appendix A. For a given readout, let $z_{\ell,t} \in \mathbb{R}^{|V|}$ denote its vocabulary-space output at layer ℓ and generation step t . For each value V_k , we define a fixed token set $\mathcal{T}_k \subset \mathcal{V}$ containing surface forms associated with that value (see Appendix B for details), following *Moral Foundations Theory* (MFT) terminology (Graham et al., 2013). These sets are fixed before evaluation and are independent of model activations. Extending them with additional value-specific terms from external lexicons (Atari et al., 2023) may improve observer sensitivity and is left for future work.

Vocabulary-space readouts are known to expose evolving intermediate predictions (nostalgebraist, 2020; Belrose et al., 2023), and prior work has shown that such projections can reveal interpretable concept structure in vocabulary space (Geva et al., 2022). Because different readout methods may produce scores with different offsets and scales, as also observed when comparing logit distributions across sources (Sun et al., 2024), we standardize each readout with respect to its vocabulary distribution at the current decoding step: $\hat{z}_{\ell,t} = \frac{z_{\ell,t} - \text{Mean}(z_{\ell,t})}{\text{Std}(z_{\ell,t})}$, where $\text{Mean}(\cdot)$ and $\text{Std}(\cdot)$ are computed over the vocabulary dimension. This places candidate readouts on a common scale without requiring learned normalization parameters or readout-specific calibration. Motivated by the bag-of-words attribute model of PPLM (Dathathri et al., 2020), which aggregates output-distribution probability mass over a curated word set to score an attribute, we introduce the *value-specific readout score* ($s_{k,\ell,t}$), which aggregates standardized vocabulary evidence for value V_k at an arbitrary intermediate layer ℓ rather than only at the final output distribution, with larger values indicating stronger value-specific evidence.

Definition 1 (Value-Specific Readout Score). Given a fixed token vocabulary ($\mathcal{T}_k$) for a targeted value V_k , layer ℓ , and generation step t , the *value-specific readout score* is defined as

$$\text{score}_{\ell,t}(V_k) := s_{k,\ell,t} = \frac{1}{|\mathcal{T}_k|} \sum_{w \in \mathcal{T}_k} \hat{z}_{\ell,t}[w]. \quad (2)$$

We map this unbounded score to a bounded value state via a sigmoid, following the use of sigmoid-transformed linear scores as bounded confidence signals for gating or scaling activation interventions in prior work (Hościłowicz et al., 2024; Nguyen et al., 2025b): $c_{k,\ell,t} = \sigma(s_{k,\ell,t}) = \frac{1}{1+\exp(-s_{k,\ell,t})}$. Thus, $c_{k,\ell,t} \in (0, 1)$, with larger values indicating stronger observed expression of V_k .

For layer ℓ and generation step t , the *intermediate-layer value observer* maps the current intermediate representation $h_{\ell,t}$ to $\mathbf{c}_{\ell,t} = [c_{1,\ell,t}, \dots, c_{K,\ell,t}]^\top \in (0, 1)^K$. The vector $\mathbf{c}_{\ell,t}$ provides a bounded estimate of the model’s current value orientation under a readout. Unlike the trained per-attribute gates used in prior sigmoid-based interventions (Hościłowicz et al., 2024; Nguyen et al., 2025b), our observer requires no probe training: $\mathbf{c}_{\ell,t}$ is computed directly from the standardized vocabulary evidence in equation 2, and depends only on the current intermediate representation, using no information from previous decoding steps. This keeps the feedback mechanism lightweight, training-free, and responsive to changes during generation.

3.3 OBSERVER-GUIDED CAUSAL MULTI-VALUE STEERING

This final stage of AIMES performs adaptive causal multi-value steering by directly intervening on the model’s hidden representations. We use *causal* here in the interventionist sense; rather than only measuring associations in representation space, AIMES modifies the hidden state along value-specific directions and evaluates the resulting change in model behavior. For a fixed readout method and intervention layer ℓ , the observer provides the current value state $\mathbf{c}_{\ell,t} = [c_{1,\ell,t}, \dots, c_{K,\ell,t}]^\top$. Let $\mathbf{g} = [g_1, \dots, g_K]^\top$, and $g_k \in \{-1, 0, +1\}$ encode the steering objective for value V_k : +1 for amplification, -1 for suppression, and 0 for no intervention. At each decoding step, AIMES uses the current observer state to determine the contribution of each requested value direction. The resulting joint intervention is

$$h'_{\ell,t} = h_{\ell,t} + \gamma \left[\sum_{k:g_k=+1} (1 - c_{k,\ell,t}) u_{k,\ell} - \sum_{k:g_k=-1} c_{k,\ell,t} u_{k,\ell} \right]. \quad (3)$$

Here, $\gamma > 0$ is the shared base steering magnitude and $u_{k,\ell}$ is the unit-normalized direction for value V_k . The observer state $c_{k,\ell,t}$ determines the value-specific steering strength: amplification is weighted by $(1 - c_{k,\ell,t})$, whereas suppression is weighted by $c_{k,\ell,t}$. The intervention therefore adapts to the currently observed value state at each decoding step. The controller is memoryless and depends only on the current readout, unlike PID-style controllers that accumulate error over time (Bharadwaj, 2025). Together, Sections 3.1-3.3 define the complete AIMES framework: value directions specify the intervention axes, the intermediate-layer readout estimates the current value state, and the controller maps these estimates to adaptive steering coefficients during generation. The full AIMES algorithm and inference procedure are provided in Appendix B.

4 EXPERIMENTAL ANALYSIS AND RESULTS

Experimental Setup and Baselines. We evaluate five instruction-tuned models spanning three families and parameter scales: Gemma-3-4B-IT and Gemma-3-12B-IT (Gemma Team et al., 2025), Qwen3-4B and Qwen3-14B (Yang et al., 2025), and Llama-3.1-8B-Instruct (Grattafiori et al., 2024). Following (Yu et al., 2026), we study the five MFT dimensions: *Care*, *Fairness*, *Loyalty*, *Authority*, and *Sanctity* (Graham et al., 2013). For each foundation, we construct 200 matched positive-negative vignette pairs to derive value-specific steering directions, with separate examples reserved for validation.

Our multi-value evaluation considers three objectives: ($\uparrow$ Care $\uparrow$ Fairness), ($\uparrow$ Loyalty $\uparrow$ Authority), and ($\uparrow$ Care $\uparrow$ Fairness $\downarrow$ Sanctity), and each objective is evaluated across ten model-specific intervention depths spanning the network. We compare AIMES with two non-adaptive baselines: (i) **Fixed Multi-Value Steering** which uses the same directions $u_{k,\ell}$, prompts, intervention layers, generation settings, and base magnitude as AIMES, but applies all requested directions simultaneously with fixed strength, $\Delta h_{\ell,t} = \gamma \sum_{k:g_k \neq 0} g_k u_{k,\ell}$, without intermediate-state feedback. This is the direct multi-value extension of contrastive activation addition (Rimsky et al., 2024) and corresponds to naive vector composition in multi-attribute steering (van der Weij et al., 2024). (ii) **Intensity-Anchor Prompt Steering** which expresses the same multi-value objective directly in the prompt (Kim et al., 2026), without modifying model activations. Because it is in-

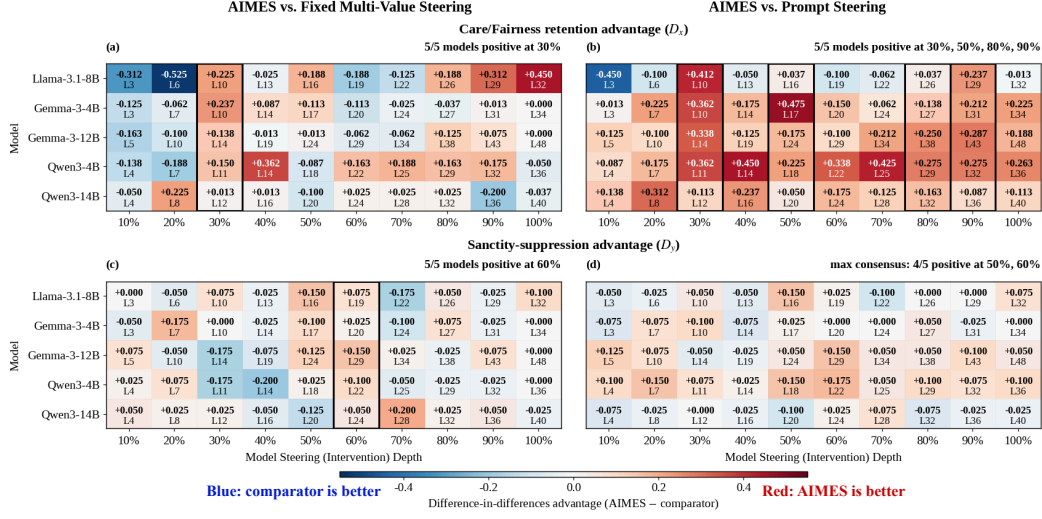


Figure 3: **Depth-wise AIMES advantage under GPT-5.6-Sol.** Panels (a,b) show Care/Fairness retention (D_x) and panels (c,d) Sanctity suppression (D_y) relative to Fixed Multi-Value Steering and Prompt Steering. Positive values favor AIMES; black outlines mark depths with unanimous positive advantage across all five models.

Model	Stage	Layer	AIMES vs. Prompt Steering				AIMES vs. Fixed Multi			
			GPT-5.6-Sol		Claude-Opus-4.8		GPT-5.6-Sol		Claude-Opus-4.8	
			D_x	D_y	D_x	D_y	D_x	D_y	D_x	D_y
Llama-3.1-8B	Early	10	55.0	51.2	48.8	55.0	60.0	52.5	55.0	51.2
	Middle	19	53.8	52.5	45.0	47.5	36.2	52.5	48.8	45.0
Gemma-3-4B	Early	10	67.5	52.5	53.8	48.8	62.5	50.0	55.0	50.0
	Middle	20	58.8	53.8	50.0	51.2	45.0	50.0	46.2	52.5
Gemma-3-12B	Early	14	60.0	51.2	55.0	55.0	48.8	46.2	50.0	47.5
	Middle	29	53.8	57.5	57.5	58.8	48.8	51.2	57.5	53.8
Qwen3-4B	Early	11	62.5	52.5	58.8	51.2	51.2	46.2	63.7	46.2
	Middle	22	65.0	56.2	51.2	50.0	60.0	55.0	55.0	50.0
Qwen3-14B	Early	12	51.2	50.0	56.2	55.0	48.8	51.2	55.0	51.2
	Middle	24	56.2	51.2	53.8	53.8	51.2	51.2	48.8	50.0

Table 1: **Prompt-level consistency across judges.** Tie-adjusted AIMES consistency (%) at early (30%) and middle (60%) depths. D_x measures Care/Fairness retention and D_y Sanctity suppression.

dependent of intervention depth, the same prompt-steered response is compared with AIMES at each tested layer. Behavioral control is evaluated using blind value-specific judgments from two independent evaluators, GPT-5.6-Sol (OpenAI, 2026) and Claude Opus-4.8 (Anthropic, 2026), applied to the same generated responses. Our main analysis reports matched prompt-level interaction contrasts and prompt-level consistency under both judges, while the full depth-wise cross-judge results are provided in Appendix D.4. The interaction contrasts isolate the effect of adding a new value constraint, whereas prompt-level consistency measures how broadly the re-sulting advantage holds across examples. Response quality is evaluated separately using the blind GPT-5.6-Sol judge; complete metric definitions are provided in Appendix D. All experiments are implemented with PyTorch and Hugging Face Transformers and run on NVIDIA A100 GPUs with Google Colab Pro+; further details are provided in Appendix C, the full code is publicly available at <https://anonymous.4open.science/status/AIMES-9A6C>.

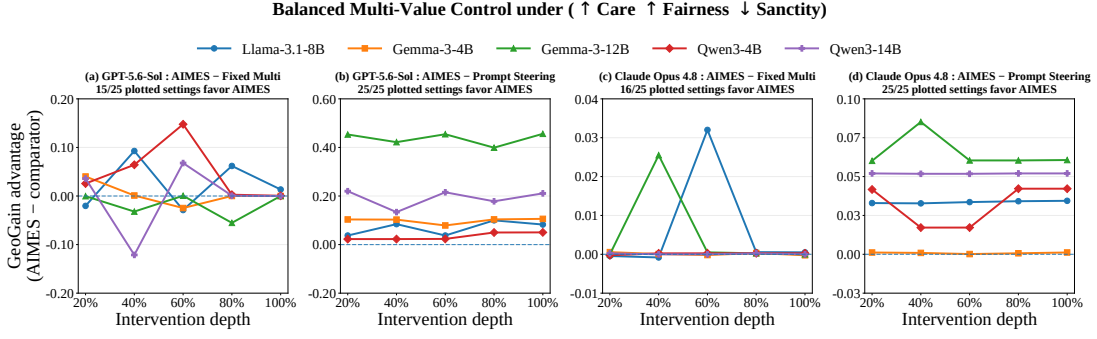


Figure 4: **GeoGain-based multi-value control evaluation.** GeoGain advantage of AIMES over Fixed Multi-Value Steering and Prompt Steering across sampled intervention depths, evaluated using two independent judges: GPT-5.6-Sol (a,b) and Claude-Opus-4.8 (c,d). In the figures, positive values favor AIMES; GeoGain measures balanced movement in $[\Delta C, \Delta F, -\Delta S]$.

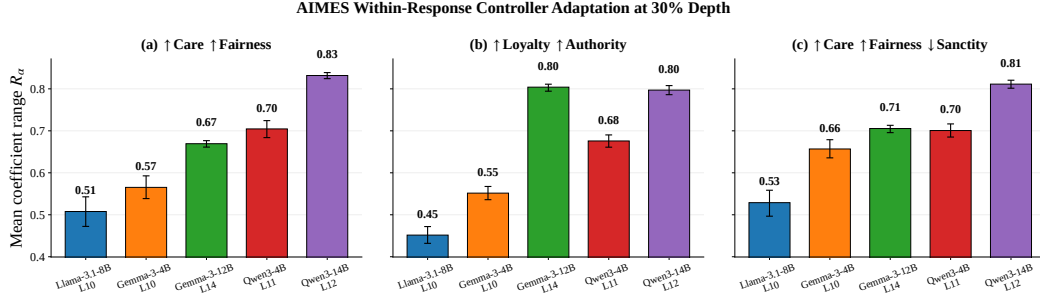


Figure 5: **Variation in AIMES steering strength within a response at 30% depth.** For each model and objective, we measure the within-response range of the steering coefficient, $R_{\alpha,k} = \max_t \alpha_{k,t} - \min_t \alpha_{k,t}$. Larger values indicate greater changes in steering strength during generation. Bars show response means and error bars denote 95% prompt-bootstrap confidence intervals.

(1) Overall Analysis of Adaptive Multi-Value Steering. Figure 3 shows that the advantage of adaptive steering depends on both intervention depth and the controlled objective. Relative to Fixed Multi-Value Steering, $D_x > 0$ for all five models at around 30% depth under GPT-5.6-Sol, a pattern that is analyzed under the independent Claude-Opus-4.8 evaluator (Appendix D.4 Figure 12). Under GPT-5.6-Sol, $D_y > 0$ for all five models at around 60% depth; however, the depth at which this Sanctity-suppression advantage concentrates shifts under Claude, indicating evaluator sensitivity. Against Prompt Steering, the Care/Fairness advantage is more persistent under GPT-5.6-Sol, with $D_x > 0$ for all five models at approximately 30%, 50%, 80%, and 90% depth, while the strongest D_y consensus reaches four of five models near 50%. Overall, the depth localization of the Care/Fairness-retention advantage is more robust across evaluators than that of the Sanctity-suppression advantage.

(2) Analyzing Prompt-Level Consistency. Table 1 complements the mean interaction contrasts in Figure 3 by measuring how broadly each advantage is distributed across prompts. Against Prompt Steering, GPT-5.6-Sol yields majority D_x consistency for all five models at both displayed depths, whereas the Claude results are more heterogeneous. Relative to Fixed Multi-Value Steering, consistency varies more strongly across models, depths, and evaluators. Thus, the mean interaction contrasts characterize the average advantage, while prompt-level consistency indicates how broadly that advantage holds across examples. Full results across all ten depths are provided in Appendix D.1.

(3) Evaluating Balanced Multi-Value Control. Figure 4 evaluates joint control of ($\uparrow$ Care $\uparrow$ Fairness $\downarrow$ Sanctity) using GeoGain (Ghasemi et al., 2026). Relative to Fixed Multi-Value Steering, the advantage of AIMES remains model- and depth-dependent, with positive GeoGain differences in 15/25 plotted settings under GPT-5.6-Sol and 16/25 under Claude-Opus-4.8. In contrast, AIMES achieves higher GeoGain than Prompt Steering in all 25 plotted settings under both judges. Thus, the balanced-control advantage is consistent relative to prompt-based steering, while the comparison with fixed joint steering remains dependent on model and intervention depth. Full layer-wise results are provided in Appendix D (Figure 10 and Figure 13).

(4) Within-Response Steering Variation. Compared to fixed activation steering, AIMES updates its intervention coefficients during generation using the online value readout. We measure this variation with the coefficient range ($R_{\alpha,k} = \max_t \alpha_{k,t} - \min_t \alpha_{k,t}$). Figure 5 shows the mean range at the representative 30% depth. Across all five models and three objectives, the ranges are consistently nonzero (0.45-0.83), showing that the steering strength varies within responses rather than remaining fixed. Full results across all ten sampled depths are provided in Appendix D.3.

5 CONCLUSION

We introduced AIMES, a closed-loop framework for adaptive activation-level steering of multiple human values using intermediate-layer vocabulary readouts as online observers. Across five models from three families, we find that multi-value controllability depends strongly on both the steering objective and intervention depth. AIMES provides depth-dependent advantages over fixed joint steering and, at favorable depths, improves joint-control performance relative to prompt-based steering, including for objectives that simultaneously promote and suppress values. These gains are achieved with smaller realized activation-space interventions than Fixed Multi-Value Steering while maintaining comparable response quality. Within the Moral Foundations Theory values and model scales studied here, AIMES offers a lightweight approach to adaptive multi-value control and motivates future work on broader value spaces, richer observers, and geometry-aware controllers.

AI USE STATEMENT

Generative AI tools were used to assist with synthetic contrastive-vignette generation, code debugging, and manuscript polishing and editing. All code, analyses, experimental decisions, and manuscript content were reviewed and verified by the authors. Synthetic data generated with AI tools were subjected to the validation and filtering procedures described in Appendix C.2. The authors take full responsibility for the final content of the paper, including all reported results, claims, and artifacts produced with the aid of generative AI.

ETHICS STATEMENT

This work studies inference-time control of value-related behavior in language models. While such methods may support more adaptable model behavior, they could also be used to impose particular value specifications or manipulate model outputs in undesirable ways. Our experiments focus on the five Moral Foundations Theory (MFT) dimensions and should not be interpreted as providing a complete or universal representation of human values. The study uses synthetic and publicly available evaluation data and does not involve direct human-subject experimentation.

REPRODUCIBILITY STATEMENT

We provide the value-direction construction, observer definition, adaptive controller, evaluation metrics, and experimental setup in the main paper. The appendix further documents the contrastive-vignette generation procedure, model-specific intervention layers, complete inference algorithm, observer ablations, value-direction geometry, and full layer-wise results. An anonymous code repository containing the implementation and supporting artifacts is linked in the experimental section.

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APPENDIX

The Appendix is organized as follows:

- A Background and Preliminaries
- B AIMES Algorithm
- C Detailed Experimental Setup
- D Additional Experimental Results

A BACKGROUND AND PRELIMINARIES

Activation-Space Control and Direction Composition. Activation steering modifies model behavior at inference time by perturbing hidden representations along directions associated with desired concepts or behaviors (Turner et al., 2023; Zou et al., 2023; Li et al., 2023). For human values, (Yu et al., 2026) construct layer-specific contrastive directions for the five Moral Foundations Theory dimensions (Graham et al., 2013) and show that interventions along these directions can shift foundation-relevant behavior. Their analysis also reveals cross-value effects, indicating that moral-value directions are not independent.

This issue is closely related to the broader problem of composing multiple activation directions. Methods such as MAT-Steer (Nguyen et al., 2025b), ORBIT (Ghasemi et al., 2026), MSRS (Jiang et al., 2025), and K-Steering (Oozeer et al., 2025) address interference among jointly controlled attributes using mechanisms including gating, orthogonalization, subspace decomposition, rotation, and nonlinear direction construction. These methods build on evidence that naive vector addition can become unreliable when several attributes are controlled simultaneously (van der Weij et al., 2024). In most cases, however, the resulting intervention rule is determined before generation and remains fixed during decoding.

Multi-Value Control. Recent work considers multiple human values through several complementary mechanisms. VALUEFLOW (Kim et al., 2026) uses natural-language value specifications to study how multiple value intensities interact, showing that their effects need not compose additively. VISPA (Zheng et al., 2026) combines separately generated value-conditioned outputs for pluralistic alignment, while (Yu et al., 2026) intervene on moral-foundation directions individually. Concurrently, (Pan et al., 2026) study joint activation-level multi-value steering and account for direction interference through a static geometry-based correction.

These works highlight two distinct challenges in multi-value control: *geometric interference* among value directions and *state dependence* during generation. The former concerns how several directions interact when applied jointly; the latter concerns whether their relative strengths should remain fixed as the model’s internal state evolves. AIMES focuses on the second problem, using online feedback to adapt the contributions of jointly applied value directions during decoding.

Feedback-Based Activation Steering. A separate line of work treats activation steering as a feedback-control problem. (Nguyen et al., 2025a) distinguish fixed-strength open-loop steering from controllers that adjust intervention magnitude according to the difference between desired and observed concept states. (Bharadwaj, 2025) use PID-style feedback with a chunk-level classifier during LLM reasoning, while (Prokopiou et al., 2026) study feedback-controlled multi-attribute steering in symbolic music generation. (Skifstad et al., 2026) instead approximate layer-wise computation as a locally linear dynamical system and derive interventions through linear-quadratic regulation.

These approaches illustrate that intervention strength can be adjusted online, but their feedback signals are typically obtained from a trained classifier, probe, or fitted dynamical model. Our setting instead uses an intermediate-layer vocabulary readout directly as the feedback signal, allowing value-specific intervention strengths to be updated without training a separate value-state estimator.

Layer-Wise Vocabulary Readouts. Several methods map intermediate transformer representations into vocabulary space to make hidden states more interpretable. The *Logit Lens* (nostalgebraist, 2020) directly applies the model’s unembedding matrix to an intermediate residual-stream activation

$$z_{\ell,t}^{\text{LL}} = W_U \text{norm}(h_{\ell,t}), \quad (4)$$

effectively treating the intermediate state as if it were already expressed in final-layer output coordinates. The *Tuned Lens* (Belrose et al., 2023) instead learns a layer-specific affine transformation,

$$z_{\ell,t}^{\text{TL}} = W_U \text{norm}(A_\ell h_{\ell,t} + b_\ell), \quad (5)$$

to better align intermediate activations with the model’s final output space. J-Lens (Gurnee et al., 2026) takes a different approach, using a layer-specific average Jacobian $J_\ell \in \mathbb{R}^{d \times d}$ to approximate how information at layer ℓ propagates toward the final representation:

$$z_{\ell,t}^J = W_U \text{norm}(J_\ell h_{\ell,t}). \quad (6)$$

The corresponding token directions form the layer-specific dictionary $D_\ell = (W_U J_\ell)^\top \in \mathbb{R}^{d \times |\mathcal{V}|}$, providing a vocabulary-aligned view of the residual stream.

Beyond interpretability, intermediate-layer vocabulary readouts have recently been connected to activation steering. (Billa, 2026) show that Logit-Lens-based accessibility can help predict both concept steerability and effective intervention depth across models and concept families. However, representational visibility does not necessarily coincide with steerability: (Nadaf, 2026) show that effective steering directions can remain weakly exposed or absent under the Logit Lens. This distinction is particularly relevant to AIMES, where the readout is used not only to inspect the intermediate representation, but also as the feedback signal that determines the strength of the subsequent intervention. We therefore compare candidate readouts as online observers under the same steering setup in Appendix C.3.

We briefly review the two lines of work most closely related to our setting: J-lens representations and moral-value tracing in LLM activation space. J-Lens (Gurnee et al., 2026) maps intermediate residual-stream activations into vocabulary-aligned coordinates while accounting for how information propagates through the remaining layers of the model. Let $h_{\ell,t} \in \mathbb{R}^d$ denote the residual-stream activation at layer ℓ and token position t , and let $W_U \in \mathbb{R}^{|\mathcal{V}| \times d}$ denote the unembedding matrix. J-Lens introduces a layer-specific average Jacobian

$$J_\ell = \mathbb{E}_{\text{prompt}, t, t' \geq t} \left[\frac{\partial h_{L,t'}}{\partial h_{\ell,t}} \right], \quad (7)$$

which provides a first-order approximation of how an intermediate representation influences later final-layer representations. The corresponding vocabulary-aligned readout is

$$z_{\ell,t}^J = W_U \text{norm}(J_\ell h_{\ell,t}). \quad (8)$$

AIMES uses this intermediate vocabulary-space signal to construct the value-specific observer scores defined in Section 3.2.

Moral-Value Representations in LLMs. (Yu et al., 2026) study how the five Moral Foundations Theory dimensions: *Care*, *Fairness*, *Loyalty*, *Authority*, and *Sanctity* (Graham et al., 2013) are represented in LLM residual streams and whether these representations can be behaviorally influenced through intervention. For a target foundation V_k , they construct a layer-specific direction from the difference between mean activations of a positive set $\mathcal{E}_k^+$ and a contrast set $\mathcal{E}_k^-$:

$$u_{k,\ell}^{\text{raw}} = \frac{1}{|\mathcal{E}_k^+|} \sum_{i \in \mathcal{E}_k^+} \tilde{h}_\ell^{(i)} - \frac{1}{|\mathcal{E}_k^-|} \sum_{j \in \mathcal{E}_k^-} \tilde{h}_\ell^{(j)}, \quad (9)$$

and normalizing that, we obtain:

$$u_{k,\ell} = \frac{u_{k,\ell}^{\text{raw}}}{\|u_{k,\ell}^{\text{raw}}\|_2} \quad (10)$$

where $\tilde{h}_\ell^{(i)}$ denotes the residual-stream representation at the final input-token position. The resulting $u_{k,\ell} \in \mathbb{R}^d$ defines a layer-specific axis associated with the target moral foundation.

The authors further analyze these directions using pretrained sparse autoencoders (SAEs), identifying decoder features aligned with $u_{k,\ell}$ and interpreting them through their top-activating contexts. They then test the behavioral relevance of the learned directions through inference-time interventions of the form

$$h'_{\ell,t} = h_{\ell,t} + \alpha_{\ell} u_{k,\ell}. \quad (11)$$

These results provide evidence that moral-foundation information is organized along distinguishable layer-wise activation directions whose manipulation can shift foundation-relevant behavior. AIMES builds on this contrastive-direction framework, but uses matched positive-negative poles for each foundation and extends the setting from single-value interventions to adaptive joint multi-value control.

B AIMES ALGORITHM

AIMES: Inference-Time Adaptive Multi-Value Steering. For each generation run, AIMES fixes an intervention layer ℓ , readout r , and base steering magnitude γ . The intervention layer is varied across experiments to study depth effects, but remains fixed within a response. Every N decoding steps, the selected readout estimates the current expression of each controlled value and updates the corresponding adaptive coefficients. We use $N=1$ in all main experiments; $N > 1$ reduces readout cost. These coefficients determine the relative contribution of each requested value direction until the next controller update. We use a fixed value-specific token set $\mathcal{T}_k$ for each moral foundation, with ten surface-form tokens per value. The complete token sets are listed in Table 2. Because each coef-

Value	Observer token set $\mathcal{T}_k$
Care	care, compassion, kindness, empathy, mercy, protection, welfare, nurture, support, helping
Fairness	fairness, justice, equality, equity, impartiality, reciprocity, rights, fair, equitable, unbiased
Loyalty	loyalty, fidelity, solidarity, allegiance, commitment, devotion, unity, belonging, fellowship, dedication
Authority	authority, respect, duty, order, obedience, hierarchy, leadership, discipline, legitimacy, tradition
Sanctity	purity, sanctity, sacred, holiness, dignity, reverence, virtue, innocence, cleanliness, sacredness

Table 2: **Value-specific observer token sets.** Each set $\mathcal{T}_k$ contains ten fixed surface-form tokens associated with the corresponding Moral Foundations Theory value and is used to compute the value-specific readout score.

cient is computed from the observed state of its corresponding value while all requested directions are applied jointly, the same mechanism naturally supports both single-value and multi-value objectives. In our multi-value experiments, AIMES is compared with Fixed Multi-Value Steering, which uses the same value directions, intervention layers, generation settings, and base magnitude γ but fixed coefficients, and with Intensity-Anchor Prompt Steering, which expresses the same objective through the input prompt without modifying model activations.

C DETAILED EXPERIMENTAL SETUP

C.1 CONSTRUCTION AND OVERVIEW

Value-Direction Construction. We study the five Moral Foundations Theory dimensions: Care, Fairness, Loyalty, Authority, and Sanctity. For each foundation, we construct 200 matched positive-negative vignette pairs corresponding to Care/Harm, Fairness/Cheating, Loyalty/Betrayal, Authority/Subversion, and Sanctity/Degradation, and use them to derive value-specific steering directions. Each pair describes the same underlying setting while reversing the relevant moral orientation, reducing variation due to topic, actors, and writing style.

For each model and transformer layer, we extract the final-token hidden representation of the positive and negative examples and compute the normalized contrastive mean-difference direction described

Algorithm 1 AIMES: Adaptive Intervention for Multi-Value Evaluation and Steering

Input: Model M , prompt x_i , intervention layer ℓ , value directions $\{u_{k,\ell}\}_{k=1}^K$, steering objective g , token sets $\{\mathcal{T}_k\}_{k=1}^K$, readout r , base steering magnitude γ

Output: Generated response y_i

```
1: for  $t = 1, \dots, T_i$  do
2:   Compute the pre-intervention hidden state  $h_{\ell,t}$ 
3:    $z_{\ell,t} \leftarrow r(h_{\ell,t})$ 
4:   Standardize  $z_{\ell,t}$  as  $\hat{z}_{\ell,t} = \frac{z_{\ell,t} - \text{Mean}(z_{\ell,t})}{\text{Std}(z_{\ell,t})}$  (highlighted in Section 3.2)
5:   for each  $k$  such that  $g_k \neq 0$  do
6:     Compute  $s_{k,\ell,t}$  using equation 2
7:      $c_{k,\ell,t} \leftarrow \sigma(s_{k,\ell,t})$ 
8:   end for
9:   Apply the joint adaptive intervention in equation 3 to obtain  $h'_{\ell,t}$ 
10:  Continue the forward pass from  $h'_{\ell,t}$  and generate the next token
11: end for
12: return  $y_i$ 
```

in Section 3.1. Direction-construction examples are kept disjoint from all prompts used for behavioral evaluation. The same precomputed directions are used by Fixed Multi-Value Steering, and AIMES, ensuring that differences between methods arise from the intervention strategy rather than from different direction estimates.

Multi-Value Evaluation Data. For the main multi-value experiments, we construct a separate evaluation set of 100 held-out MFRC prompts, balanced across the five foundations (20 prompts per foundation). For each steering objective, we retain only prompts whose source foundation is not one of the values being directly controlled. This avoids evaluating a multi-value intervention on prompts that are already explicitly associated with one of its target foundations.

Under this filtering, the two-value objectives ($\uparrow$ Care $\uparrow$ Fairness) and ($\uparrow$ Loyalty $\uparrow$ Authority) each retain 60 prompts, corresponding to the three non-target foundations. The three-value competing objective ($\uparrow$ Care $\uparrow$ Fairness $\downarrow$ Sanctity) retains 40 prompts from the two remaining foundations, Loyalty and Authority. Each objective is evaluated at all ten model-specific intervention depths using the same prompts for AIMES, Fixed Multi-Value Steering, and Prompt Steering.

For the matched interaction analysis comparing ($\uparrow$ Care $\uparrow$ Fairness $\downarrow$ Sanctity) with ($\uparrow$ Care $\uparrow$ Fairness), we restrict both objectives to the same 40 prompts. Thus, the D_x , D_y , and prompt-level consistency analyses use $n = 40$ matched prompts per model-layer-comparator setting.

Intervention Layers. Since the evaluated architectures contain different numbers of transformer layers, we compare intervention locations using normalized depth

$$d_\ell = \frac{\ell}{L},$$

where L is the number of transformer layers. For each model, we evaluate ten approximately evenly spaced depths spanning the network. The exact layer indices are reported in Table 3. The intervention layers are selected a priori to approximately span the network depth; no layer is selected post hoc based on behavioral performance. The complete depth sweep is used to characterize how steering behavior changes through the network.

Multi-Value Steering. The multi-value experiments compare AIMES with two non-adaptive baselines: Fixed Multi-Value Steering and Intensity-Anchor Prompt Steering. Fixed Multi-Value Steering jointly applies all requested value directions at the same intervention layer using fixed coefficients, while AIMES uses the same prompts, value directions, intervention layers, generation settings, and base magnitude but replaces the fixed coefficients with the observer-dependent weights defined in Section 3.3. Prompt Steering specifies the same multi-value objective directly in the input prompt without modifying model activations. Unsteered responses are used as matched references when computing value-score changes. For each AIMES generation, the intervention layer, observer,

Model	No. Layers	Intervention Layers
Gemma-3-4B-IT	34	3, 7, 10, 14, 17, 20, 24, 27, 31, 34
Gemma-3-12B-IT	48	5, 10, 14, 19, 24, 29, 34, 38, 43, 48
Qwen3-4B	36	4, 7, 11, 14, 18, 22, 25, 29, 32, 36
Qwen3-14B	40	4, 8, 12, 16, 20, 24, 28, 32, 36, 40
Llama-3.1-8B-Instruct	32	3, 6, 10, 13, 16, 19, 22, 26, 29, 32

Table 3: Transformer layers used in the depth sweep. The layer indices are chosen to approximately cover normalized depths from 0.1 to 1.0 for each architecture.

and base magnitude are fixed, while the value-specific coefficients are recomputed at every decoding step ($N = 1$ in all main experiments). The controller is memoryless and retains no accumulated state across decoding steps.

Metrics and Statistical Analysis. All method comparisons are performed on matched prompts. Our primary analysis uses matched interaction contrasts to measure how adding a value constraint changes control relative to a shared base objective, together with prompt-level consistency to characterize how broadly the effect holds across examples. We additionally use GeoGain (Ghasemi et al., 2026) to evaluate balanced movement across all requested values, and separately report blind response quality and realized activation-space perturbation. Pointwise bootstrap confidence intervals are used where reported. Complete metric definitions and statistical details are provided in Appendix D.

C.2 CONTRASTIVE VIGNETTE GENERATION



Figure 6: **Value-specific steering direction construction.** Value examples are selected, converted into contrastive pairs, verified, and used to extract model and value-specific activation directions.

Independent semantic validity check. We have generated the contrastive value-specific and model-specific directions as shown in Figure 6. As a preliminary sanity check, we evaluated whether each generated foundation-specific vignette set was semantically distinguishable from the Social Norm reference set using three independent sentence-embedding models: BGE-large-en-v1.5, E5-large-v2, and all-MPNet-base-v2. For foundation V_k and embedding model m , we computed

$$\Delta_k^{(m)} = \mathbb{E}_{x \in \mathcal{D}_k} [\cos(e_m(x), e_m(d_k))] - \mathbb{E}_{x \in \mathcal{D}_{\text{SN}}} [\cos(e_m(x), e_m(d_k))], \quad (12)$$

where $e_m(\cdot)$ denotes the representation produced by embedding model m , and d_k denotes the textual definition of foundation V_k . A positive $\Delta_k^{(m)}$ therefore indicates that the foundation-specific vignette set is, on average, more semantically aligned with its intended foundation definition than the Social Norm reference set. All five foundations yielded positive contrasts under all three embedding models, indicating consistent set-level separation from the Social Norm distribution.

Averaged across embedding models, the observed differences were 0.063 for Care, 0.052 for Fairness, 0.060 for Loyalty, 0.035 for Authority, and 0.078 for Sanctity, with Authority showing the weakest separation and Sanctity the strongest. Because these embedding models are general-purpose semantic encoders rather than Moral Foundations Theory classifiers, we use this analysis only as a corpus-level sanity check. This foundation-versus-Social-Norm comparison also provides a semantic counterpart to the contrastive construction used in prior moral-tracing work (Yu et al., 2026), while the matched positive-negative corpus described separately is used to construct the bidirectional steering directions employed by AIMES. The primary validation of these directions is subsequently performed in the target models’ activation space.

Foundation	$\overline{\Delta}_k$	Positive Models
Care	0.063	3/3
Fairness	0.052	3/3
Loyalty	0.060	3/3
Authority	0.035	3/3
Sanctity	0.078	3/3

Table 4: Independent semantic sanity check of the generated foundation-specific vignette sets. $\overline{\Delta}_k$ denotes the mean foundation-versus-Social-Norm cosine-similarity (Salton et al., 1975) difference averaged across BGE-large-en-v1.5, E5-large-v2, and all-MPNet-base-v2.

Matched Positive-Negative Contrastive Vignette Generation. For the primary AIMES steering directions, we construct a second synthetic corpus consisting of matched positive–negative vignette pairs for each of the five Moral Foundations Theory dimensions. In contrast to the foundation-versus-Social-Norm corpus used for reproduction and representation-level comparison, this corpus is designed specifically to recover a bidirectional moral axis suitable for activation steering.

For each moral foundation V_k , we define the corresponding positive and negative poles according to the standard Moral Foundations Theory formulation:

$$\text{Care} \leftrightarrow \text{Harm}, \quad (13)$$

$$\text{Fairness} \leftrightarrow \text{Cheating}, \quad (14)$$

$$\text{Loyalty} \leftrightarrow \text{Betrayal}, \quad (15)$$

$$\text{Authority} \leftrightarrow \text{Subversion}, \quad (16)$$

$$\text{Sanctity} \leftrightarrow \text{Degradation}. \quad (17)$$

These paired dimensions follow the canonical virtue–vice structure of Moral Foundations Theory (Graham et al., 2011; 2013). For each foundation, we generate $N = 200$ matched contrastive pairs,

$$\mathcal{D}_k^\pm = \left\{ \left(x_{k,i}^+, x_{k,i}^- \right) \right\}_{i=1}^{200}, \quad (18)$$

where $x_{k,i}^+$ expresses the positive pole of the foundation and $x_{k,i}^-$ expresses its corresponding negative pole. Equivalently, these matched pairs induce positive and negative example sets

$$\mathcal{D}_k^+ = \left\{ x_{k,i}^+ \right\}_{i=1}^{200}, \quad \mathcal{D}_k^- = \left\{ x_{k,i}^- \right\}_{i=1}^{200}. \quad (19)$$

Across the five foundations, this produces 1,000 matched pairs, or 2,000 individual vignettes.

The two members of each pair are generated jointly rather than independently. The generator is instructed to preserve, as closely as possible, the same actors, setting, underlying event, grammatical structure, level of detail, and approximate length across the positive and negative versions. The principal difference between the two members should be the moral polarity associated with the target foundation. This matched construction reduces variation due to topic, setting, lexical content, and writing style, making the resulting activation difference more directly attributable to the target moral dimension.

For example, a Care/Harm pair may take the form

Care: A student stopped to comfort a classmate who was crying after being mocked.

Harm: A student joined the others in mocking a classmate who was already crying.

Similarly, a Fairness/Cheating pair may contrast returning money accidentally overpaid by a customer with intentionally keeping the excess amount. The specific wording is not fixed; rather, the generation procedure enforces structural similarity within each pair while varying scenarios across the corpus.

To promote contextual diversity, the same twelve generation-time social settings used in the foundation-versus-Social-Norm corpus are retained:

1. family and household,
2. friends and interpersonal relationships,
3. school and education,
4. workplace,
5. community and neighborhood,
6. groups, clubs, and organizations,
7. public and service interactions,
8. online and digital interactions,
9. sports and recreation,
10. health and caregiving,
11. travel and transportation, and
12. social gatherings.

These settings are used only as diversity controls and are not treated as moral labels or theoretically defined context categories. Generation is approximately balanced across the twelve contexts, yielding roughly 16–17 pairs per context for each foundation. Each generated record stores the pair identifier, target foundation, diversity context, positive vignette, negative vignette, generator identifier, and generation metadata. The resulting data therefore have the conceptual structure

$$(\text{pair_id}, \text{foundation}, \text{context}, \text{positive_text}, \text{negative_text}). \quad (20)$$

To construct the layer-wise activation direction, we first compute the mean hidden representation of the positive and negative sets at layer ℓ :

$$\mu_{k,\ell}^+ = \frac{1}{N} \sum_{i=1}^N h_\ell(x_{k,i}^+), \quad \mu_{k,\ell}^- = \frac{1}{N} \sum_{i=1}^N h_\ell(x_{k,i}^-). \quad (21)$$

We then define the normalized positive–negative direction as

$$u_{k,\ell}^{\text{PN}} = \frac{\mu_{k,\ell}^+ - \mu_{k,\ell}^-}{\|\mu_{k,\ell}^+ - \mu_{k,\ell}^-\|_2}. \quad (22)$$

Because the two sets consist of one-to-one matched pairs of equal size, the unnormalized mean difference is equivalently

$$\mu_{k,\ell}^+ - \mu_{k,\ell}^- = \frac{1}{N} \sum_{i=1}^N [h_\ell(x_{k,i}^+) - h_\ell(x_{k,i}^-)]. \quad (23)$$

Thus, the mean-difference and pairwise-difference formulations yield the same direction. Here, $h_\ell(\cdot)$ denotes the hidden representation extracted from layer ℓ using a fixed token-selection rule. This construction follows the general contrastive mean-difference principle used in activation-steering methods (Rimsky et al., 2024).

The positive–negative corpus is kept separate from the foundation-versus-Social-Norm corpus. The latter is used primarily for reproducing and comparing against the representation structure reported in prior moral-tracing work, whereas the positive–negative corpus provides the primary bidirectional steering directions used by AIMES.

C.3 OBSERVER ABLATION: J-LENS VS. LOGIT LENS

Figure 7 compares J-Lens and Logit Lens as the feedback observer while holding the steering directions, controller, intervention layers, prompts, and generation settings fixed. J-Lens exhibits higher directional responsiveness in all five models, indicating that its internal value estimates more consistently move in the requested direction under intervention. Downstream, AIMES-JL achieves higher worst-target gain in four of five models, lower non-target interference in three of five, and higher response quality in four of five. Most behavioral confidence intervals overlap zero, indicating that the downstream advantage is not uniform across architectures; however, these results, together with the consistently stronger directional responsiveness, motivate our use of J-Lens as the default observer in the main experiments.

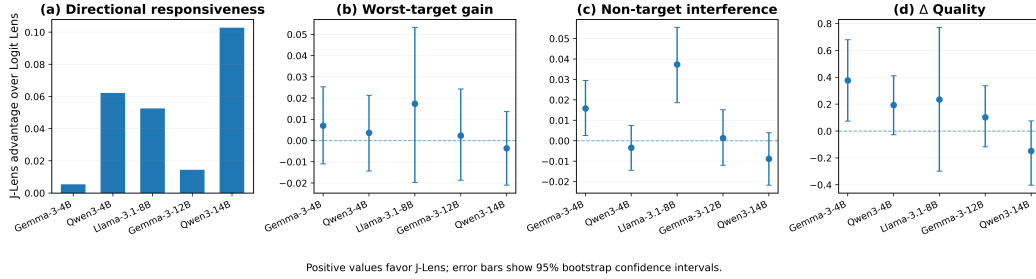


Figure 7: **Observer ablation: J-Lens versus Logit Lens.** Positive values favor J-Lens. J-Lens yields higher directional responsiveness in all five models and generally favorable downstream trends for worst-target gain, non-target interference, and response quality. Error bars in panels (b)-(d) denote 95% bootstrap confidence intervals.

C.4 LAYER-WISE GEOMETRY OF VALUE DIRECTIONS

Figure 8 shows the pairwise cosine similarity among the five value directions at representative early, middle, and late layers for all five models. The geometry is strongly model and depth-dependent. In Llama-3.1-8B-Instruct, pairwise similarities are predominantly positive and relatively stable across depth, although Loyalty remains less aligned with the other foundations. The two Qwen3 models exhibit a clearer depth-dependent transition: Loyalty is nearly orthogonal, and in some cases weakly anti-aligned, with several other directions at early layers, while the directions become more positively correlated in middle and late layers.

The Gemma models show the largest changes across depth. In Gemma-3-4B-IT, several early-layer pairs are negatively aligned, whereas the middle-layer directions become highly correlated before separating again toward later layers. Gemma-3-12B-IT similarly exhibits strong positive and negative relationships at early depth, followed by broadly positive correlations in the middle and late network. These results indicate that the interaction structure among value directions cannot be treated as fixed across either models or layers. Consequently, simultaneously applying several directions may produce substantially different geometric interactions depending on the intervention depth, motivating our evaluation of multi-value steering across multiple depths rather than selecting a single layer a priori.

AIMES achieves control with smaller realized interventions. Figure 9 compares the realized activation-space update of AIMES with Fixed Multi-Value Steering across all 150 model-layer-objective configurations. AIMES produces a smaller perturbation in every setting, with reductions ranging from approximately 8% to 87%; the pointwise 95% bootstrap confidence interval excludes zero throughout. The magnitude of the reduction varies substantially across models, layers, and objectives, indicating that the adaptive controller does not simply rescale the fixed intervention by a constant factor. Thus, the behavioral gains reported in the main text are obtained with smaller realized activation updates rather than stronger interventions.

Full Layer-Wise GeoGain Analysis. Figure 10 extends the balanced-control analysis across the full set of tested intervention depths. The comparison with Fixed Multi-Value Steering varies substantially across models and layers, confirming that the relative benefit of adaptive steering is depth-dependent. In contrast, the comparison with Prompt Steering is more consistently positive across models and depth. These results reinforce the main-text finding that balanced control under the competing objective depends strongly on intervention location, particularly when compared with fixed joint steering.

D ADDITIONAL EXPERIMENTAL RESULTS

Evaluation Metrics. We evaluate adaptive multi-value control using four complementary measures. We use a shared base steering magnitude of $\gamma = 1.0$ for all main multi-value experiments.

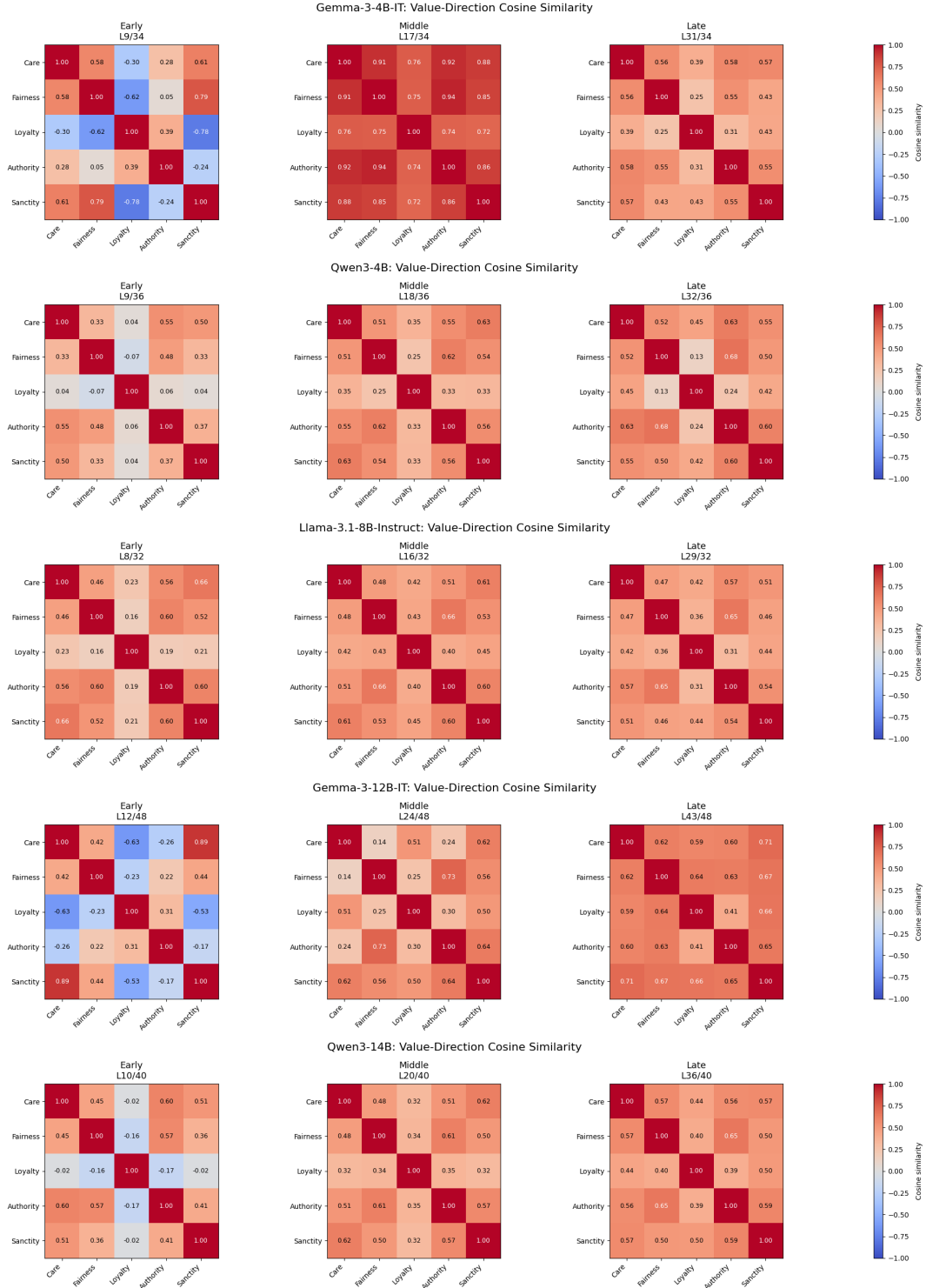


Figure 8: Pairwise cosine similarity among the five value directions across models and network depth. Each model is shown at representative early, middle, and late layers, with entries reporting $\cos(u_{i,\ell}, u_{j,\ell})$ for Care, Fairness, Loyalty, Authority, and Sanctity. Positive values indicate aligned directions, values near zero indicate approximate orthogonality, and negative values indicate opposite geometry.

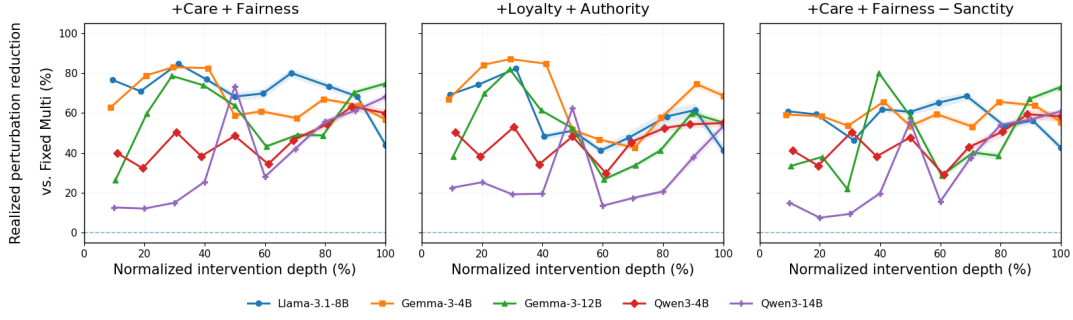


Figure 9: **Realized activation-space perturbation reduction relative to Fixed Multi-Value Steering**, $R_\ell = 100 (1 - \|\Delta h_{\text{AIMES},\ell}\|_2 / \|\Delta h_{\text{Fixed},\ell}\|_2)$, on shared prompt states. $R_\ell > 0$ in all 150 model-layer-objective settings (95% CIs excluding zero), with magnitude varying substantially by model, depth, and objective rather than reflecting a constant rescaling.

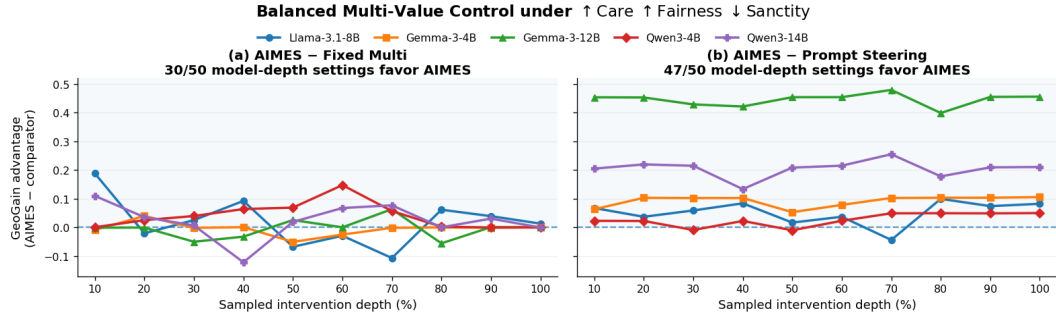


Figure 10: **GPT-5.6-Sol GeoGain advantage for balanced multi-value control under ($\uparrow$ Care $\uparrow$ Fairness $\downarrow$ Sanctity)**. GeoGain uses the geometric mean of the aligned target changes $[\Delta C, \Delta F, -\Delta S]$, with a positive sign only when all targets move in the requested directions. Curves show the prompt-paired mean (AIMES-comparator) difference across intervention depths; positive values favor AIMES.

(a) Matched Interaction Contrast. We use a matched interaction contrast, adapting the difference-in-differences formulation (Card & Krueger, 1994; Angrist & Pischke, 2009), to measure how adding a new value constraint changes control relative to a shared base objective. Let $A = (\uparrow \text{Care } \uparrow \text{Fairness } \downarrow \text{Sanctity})$ and $B = (\uparrow \text{Care } \uparrow \text{Fairness})$. For prompt i , method m , objective o , and layer ℓ , we define

$$G_{\text{CF},i}^{(m,o,\ell)} = \frac{\Delta C_i^{(m,o,\ell)} + \Delta F_i^{(m,o,\ell)}}{2}, \quad G_{\text{S},i}^{(m,o,\ell)} = -\Delta S_i^{(m,o,\ell)}, \quad (24)$$

where changes are measured relative to the corresponding unsteered response. The incremental effect of adding $\downarrow$ Sanctity is $I_x = G_{\text{CF}}^A - G_{\text{CF}}^B$ for Care/Fairness retention and $I_y = G_{\text{S}}^A - G_{\text{S}}^B$ for additional Sanctity suppression. For comparator $q \in \{\text{Fixed}, \text{Prompt}\}$, we report

$$D_x^{(q)}(\ell) = \frac{1}{n} \sum_i \left[I_{x,i}^{(\text{AIMES},\ell)} - I_{x,i}^{(q,\ell)} \right], \quad D_y^{(q)}(\ell) = \frac{1}{n} \sum_i \left[I_{y,i}^{(\text{AIMES},\ell)} - I_{y,i}^{(q,\ell)} \right]. \quad (25)$$

Positive values favor AIMES. We additionally report *prompt-level consistency*, the tie-adjusted percentage of matched prompts for which the corresponding (AIMES-comparator) incremental effect is positive.

(b) Geometric Gain (GeoGain). To measure balanced progress across all components of a composite objective, we use ORBIT-style Geometric Gain (GeoGain) (Ghasemi et al., 2026). For aligned

target changes $\tilde{\Delta}_k = g_k \Delta_k$, we compute

$$M_\epsilon = \exp\left(\frac{1}{K} \sum_{k=1}^K \log(|\tilde{\Delta}_k| + \epsilon)\right),$$

and assign $\text{GeoGain} + M_\epsilon$ only when all $\tilde{\Delta}_k > 0$, and $-M_\epsilon$ otherwise. GeoGain therefore rewards simultaneous movement in all requested directions while penalizing imbalanced control. Method comparisons use prompt-paired differences before averaging within each model–layer setting.

(c) Within-Response Controller Adaptation. Prior work on adaptive activation steering has shown the utility of varying steering intensity during inference (Scalena et al., 2024; Wang et al., 2025; Zhao et al., 2025). To quantify the realized amount of token-level adaptation in our controller, we use the range of each target-value coefficient trajectory,

$$R_{\alpha,k} = \max_t \alpha_{k,t} - \min_t \alpha_{k,t},$$

where $\alpha_{k,t}$ denotes the steering coefficient applied to value V_k at decoding step t . For each model, steering objective, and intervention depth, we average $R_{\alpha,k}$ across target values and responses. Larger values indicate greater within-response variation, whereas values near zero indicate that the corresponding coefficient remains nearly constant throughout generation. This diagnostic characterizes coefficient variation only and does not, by itself, establish whether that variation is behaviorally beneficial.

(d) Response Quality. Since, stronger behavioral control can potentially come at the cost of generation quality, we independently evaluate whether each steering method preserves the quality of the generated response. We use a blind GPT-5.6-Sol (OpenAI, 2026) judge that observes only the original prompt and generated response; the steering method, intervention layer, steering objective, and source foundation are hidden from the judge. The judge independently scores *coherence* (C), *fluency* (F), and *relevance* (R) on a four-point scale, and we summarize overall response quality as

$$Q = \frac{C + F + R}{3}.$$

We report both the raw quality scores and their changes relative to the corresponding unsteered response, $\Delta Q = Q_{\text{steered}} - Q_{\text{unsteered}}$, with analogous definitions for coherence, fluency, and relevance. Identical prompt–response pairs are judged only once and their scores are reused across duplicated conditions, such as the layer-independent Prompt Steering baseline. All quality comparisons are computed at the exact intervention layer, without averaging across layers.

D.1 ROBUSTNESS ANALYSIS

GPT-5.6-Sol: Full depth-wise prompt-level consistency.

Model	Depth	ℓ	vs. Prompt Steering		vs. Fixed Multi	
			D_x	D_y	D_x	D_y
Llama-3.1-8B	10%	3	42.5 [28.7, 56.2]	50.0 [43.8, 57.5]	41.2 [27.5, 55.0]	50.0 [43.8, 56.2]
	20%	6	51.2 [37.5, 65.0]	50.0 [41.2, 58.8]	38.7 [25.0, 53.8]	50.0 [41.2, 58.8]
	30%	10	55.0 [40.0, 70.0]	51.2 [43.8, 58.8]	60.0 [46.2, 73.8]	52.5 [47.5, 57.5]
	40%	13	50.0 [36.3, 63.7]	47.5 [40.0, 53.8]	36.2 [22.5, 51.2]	48.8 [42.5, 55.0]
	50%	16	51.2 [36.2, 66.2]	55.0 [48.8, 61.3]	51.2 [38.7, 65.0]	56.2 [48.8, 63.7]
	60%	19	53.8 [38.8, 67.5]	52.5 [43.8, 61.3]	36.2 [23.8, 50.0]	52.5 [45.0, 60.0]
	70%	22	45.0 [31.2, 58.8]	47.5 [41.2, 53.8]	50.0 [37.5, 62.5]	43.8 [36.3, 50.0]
	80%	26	50.0 [36.2, 63.8]	50.0 [41.2, 58.8]	57.5 [43.8, 71.2]	51.2 [43.8, 58.8]
	90%	29	60.0 [47.5, 72.5]	50.0 [42.5, 57.5]	58.8 [46.2, 71.2]	47.5 [40.0, 55.0]
	100%	32	48.8 [36.2, 62.5]	51.2 [42.5, 58.8]	65.0 [53.8, 76.2]	52.5 [46.2, 58.7]

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Table 5 – continued

Model	Depth	L	vs. Prompt Steering		vs. Fixed Multi	
			D_x Cons.	D_y Cons.	D_x Cons.	D_y Cons.
Gemma-3-4B	10%	3	51.2 [37.5, 65.0]	48.8 [41.2, 56.2]	40.0 [27.5, 52.5]	46.2 [38.8, 53.8]
	20%	7	56.2 [43.8, 68.8]	53.8 [46.2, 61.3]	43.8 [30.0, 57.5]	55.0 [48.8, 62.5]
	30%	10	67.5 [55.0, 78.8]	52.5 [43.8, 61.3]	62.5 [50.0, 75.0]	50.0 [43.8, 56.2]
	40%	14	61.3 [48.8, 73.8]	48.8 [42.5, 55.0]	51.2 [38.8, 63.7]	48.8 [42.5, 55.0]
	50%	17	67.5 [55.0, 78.8]	51.2 [43.8, 58.8]	53.8 [41.2, 66.2]	53.8 [48.8, 60.0]
	60%	20	58.8 [46.2, 71.2]	53.8 [46.2, 61.3]	45.0 [32.5, 57.5]	50.0 [43.8, 57.5]
	70%	24	57.5 [45.0, 70.0]	50.0 [42.5, 57.5]	50.0 [38.7, 61.3]	47.5 [42.5, 52.5]
	80%	27	53.8 [41.2, 67.5]	51.2 [45.0, 57.5]	45.0 [33.8, 56.2]	53.8 [50.0, 58.7]
	90%	31	56.2 [43.8, 68.8]	51.2 [43.8, 58.8]	48.8 [38.8, 58.7]	48.8 [45.0, 52.5]
	100%	34	58.8 [46.2, 71.2]	51.2 [45.0, 57.5]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]
Gemma-3-12B	10%	5	53.8 [40.0, 67.5]	55.0 [47.5, 63.7]	41.2 [31.2, 51.2]	51.2 [46.2, 56.2]
	20%	10	55.0 [41.2, 68.8]	55.0 [46.2, 63.7]	47.5 [36.2, 58.8]	48.8 [41.2, 56.2]
	30%	14	60.0 [46.2, 73.8]	51.2 [43.8, 60.0]	48.8 [37.5, 58.8]	46.2 [40.0, 52.5]
	40%	19	52.5 [38.8, 66.2]	50.0 [41.2, 58.8]	55.0 [43.8, 66.2]	48.8 [43.8, 53.8]
	50%	24	56.2 [41.2, 70.0]	52.5 [45.0, 60.0]	48.8 [37.5, 60.0]	55.0 [48.8, 62.5]
	60%	29	53.8 [40.0, 67.5]	57.5 [50.0, 65.0]	48.8 [37.5, 60.0]	51.2 [46.2, 56.2]
	70%	34	56.2 [42.5, 70.0]	55.0 [47.5, 62.5]	51.2 [40.0, 61.3]	51.2 [45.0, 57.5]
	80%	38	57.5 [42.5, 72.5]	52.5 [45.0, 60.0]	55.0 [46.2, 63.7]	48.8 [45.0, 52.5]
	90%	43	58.8 [45.0, 71.2]	53.8 [46.2, 61.3]	53.8 [45.0, 62.5]	52.5 [47.5, 57.5]
	100%	48	57.5 [43.8, 71.2]	52.5 [45.0, 60.0]	50.0 [50.0, 50.0]	50.0 [50.0, 50.0]
Qwen3-4B	10%	4	53.8 [40.0, 67.5]	55.0 [47.5, 62.5]	47.5 [37.5, 57.5]	51.3 [47.5, 55.0]
	20%	7	61.3 [47.5, 73.8]	56.2 [50.0, 62.5]	46.2 [35.0, 57.5]	51.2 [45.0, 57.5]
	30%	11	62.5 [48.8, 76.2]	52.5 [45.0, 60.0]	51.2 [40.0, 62.5]	46.2 [41.2, 51.3]
	40%	14	61.3 [48.8, 73.8]	51.2 [43.8, 60.0]	62.5 [53.8, 71.2]	45.0 [38.8, 50.0]
	50%	18	57.5 [43.8, 71.2]	52.5 [45.0, 60.0]	46.2 [35.0, 57.5]	51.2 [45.0, 57.5]
	60%	22	65.0 [52.5, 77.5]	56.2 [48.8, 63.7]	60.0 [50.0, 70.0]	55.0 [50.0, 61.3]
	70%	25	61.3 [48.8, 73.8]	53.8 [46.2, 61.3]	56.2 [48.8, 65.0]	48.8 [45.0, 52.5]
	80%	29	61.3 [47.5, 73.8]	53.8 [46.2, 61.3]	52.5 [43.8, 61.3]	50.0 [46.2, 53.8]
	90%	32	57.5 [43.8, 71.2]	52.5 [45.0, 58.8]	55.0 [50.0, 61.3]	48.8 [46.2, 50.0]
	100%	36	60.0 [47.5, 72.5]	53.8 [46.2, 61.3]	47.5 [42.5, 52.5]	50.0 [50.0, 50.0]
Qwen3-14B	10%	4	55.0 [41.2, 67.5]	48.8 [40.0, 57.5]	47.5 [36.3, 58.7]	52.5 [50.0, 56.2]
	20%	8	63.8 [50.0, 76.2]	48.8 [40.0, 57.5]	53.8 [41.2, 66.3]	50.0 [45.0, 55.0]
	30%	12	51.2 [37.5, 65.0]	50.0 [41.2, 58.8]	48.8 [38.7, 58.7]	51.2 [46.2, 56.2]
	40%	16	55.0 [41.2, 68.8]	50.0 [41.2, 58.8]	55.0 [43.8, 66.2]	47.5 [42.5, 52.5]
	50%	20	50.0 [36.3, 63.7]	48.8 [40.0, 57.5]	45.0 [33.8, 57.5]	48.8 [43.8, 53.8]
	60%	24	56.2 [43.8, 67.5]	51.2 [43.8, 60.0]	51.2 [40.0, 62.5]	51.2 [46.2, 56.2]
	70%	28	56.2 [42.5, 70.0]	52.5 [43.8, 61.3]	52.5 [42.5, 62.5]	53.8 [50.0, 58.7]
	80%	32	58.8 [45.0, 71.2]	48.8 [40.0, 57.5]	48.8 [40.0, 58.8]	51.3 [47.5, 55.0]
	90%	36	55.0 [42.5, 68.8]	50.0 [41.2, 58.8]	40.0 [32.5, 47.5]	51.2 [50.0, 53.8]
	100%	40	55.0 [42.5, 67.5]	50.0 [41.2, 58.8]	47.5 [43.8, 50.0]	48.8 [46.2, 50.0]

Table 4: Prompt-level consistency across all intervention depths. Tie-adjusted AIMES consistency (%) with pointwise 95% prompt-bootstrap confidence intervals in brackets ($n = 40$ matched prompts per cell).

Claude-Opus-4.8: Full depth-wise prompt-level consistency. Table 6 complements the Claude mean interaction contrasts in Figure 12 by showing how consistently the corresponding advantage occurs across individual prompts at every tested intervention depth. Each entry reports the tie-adjusted percentage of matched prompts for which AIMES obtains a larger D_x or D_y interaction effect than the corresponding comparator. Values above 50% therefore indicate that the AIMES advantage occurs for a majority of prompts. The results show that prompt-level consistency varies across models, objectives, and intervention depths, while positive-majority consistency appears across a broad range of settings for both comparison methods.

Model	Depth	ℓ	vs. Prompt Steering		vs. Fixed Multi	
			D_x	D_y	D_x	D_y
Llama-3.1-8B	10%	3	31.2 [21.2, 42.5]	48.8 [41.2, 56.2]	45.0 [33.8, 56.2]	50.0 [43.8, 57.5]
	20%	6	43.8 [31.2, 56.2]	48.8 [41.2, 56.2]	45.0 [33.8, 56.2]	47.5 [41.2, 53.8]
	30%	10	48.8 [36.3, 61.3]	55.0 [47.5, 62.5]	55.0 [43.8, 66.2]	51.2 [45.0, 57.5]
	40%	13	42.5 [31.2, 53.8]	55.0 [48.8, 62.5]	52.5 [38.8, 66.2]	56.2 [51.2, 61.3]
	50%	16	42.5 [31.2, 53.8]	51.2 [43.8, 58.8]	53.8 [42.5, 65.0]	52.5 [46.2, 58.8]
	60%	19	45.0 [32.5, 57.5]	47.5 [40.0, 55.0]	48.8 [37.5, 58.8]	45.0 [37.5, 51.2]
	70%	22	41.2 [28.7, 53.8]	52.5 [46.2, 60.0]	53.8 [42.5, 65.0]	50.0 [43.8, 56.2]
	80%	26	43.8 [32.5, 55.0]	50.0 [42.5, 57.5]	41.2 [31.2, 51.2]	47.5 [41.2, 53.8]
	90%	29	51.2 [38.8, 63.7]	53.8 [46.2, 61.3]	56.2 [46.2, 65.0]	51.2 [46.2, 56.2]
	100%	32	50.0 [37.5, 62.5]	51.2 [45.0, 57.5]	55.0 [46.2, 63.7]	50.0 [45.0, 55.0]
Gemma-3-4B	10%	3	58.8 [48.8, 68.8]	48.8 [42.5, 53.8]	52.5 [42.5, 62.5]	51.2 [46.2, 56.2]
	20%	7	53.8 [43.8, 63.7]	48.8 [42.5, 53.8]	56.2 [48.8, 63.7]	48.8 [43.8, 53.8]
	30%	10	53.8 [43.8, 62.5]	48.8 [42.5, 55.0]	55.0 [47.5, 62.5]	50.0 [45.0, 55.0]
	40%	14	52.5 [42.5, 62.5]	48.8 [43.8, 53.8]	47.5 [38.7, 56.2]	48.8 [43.8, 53.8]
	50%	17	52.5 [43.8, 61.3]	47.5 [41.2, 53.8]	48.8 [40.0, 56.2]	51.2 [46.2, 56.2]
	60%	20	50.0 [41.2, 58.8]	51.2 [45.0, 57.5]	46.2 [38.7, 53.8]	52.5 [46.2, 58.8]
	70%	24	51.2 [40.0, 62.5]	48.8 [43.8, 53.8]	51.2 [41.2, 61.3]	50.0 [46.2, 53.8]
	80%	27	53.8 [45.0, 62.5]	51.2 [46.2, 56.2]	56.2 [48.8, 63.7]	46.2 [41.2, 50.0]
	90%	31	53.8 [45.0, 62.5]	48.8 [42.5, 53.8]	53.8 [47.5, 60.0]	48.8 [46.2, 50.0]
	100%	34	53.8 [45.0, 62.5]	48.8 [43.8, 53.8]	47.5 [41.2, 53.8]	50.0 [50.0, 50.0]
Gemma-3-12B	10%	5	57.5 [47.5, 67.5]	55.0 [48.8, 62.5]	50.0 [42.5, 57.5]	48.8 [45.0, 52.5]
	20%	10	52.5 [42.5, 62.5]	55.0 [48.8, 62.5]	48.8 [40.0, 57.5]	51.3 [47.5, 55.0]
	30%	14	55.0 [45.0, 65.0]	55.0 [50.0, 61.3]	50.0 [42.5, 57.5]	47.5 [43.8, 50.0]
	40%	19	53.8 [43.8, 63.7]	55.0 [50.0, 61.3]	48.8 [40.0, 56.2]	50.0 [45.0, 55.0]
	50%	24	55.0 [45.0, 65.0]	53.8 [47.5, 60.0]	51.2 [46.2, 56.2]	50.0 [46.2, 53.8]
	60%	29	57.5 [46.2, 67.5]	58.8 [52.5, 66.2]	57.5 [50.0, 65.0]	53.8 [50.0, 58.7]
	70%	34	53.8 [43.8, 63.7]	52.5 [46.2, 60.0]	51.2 [43.8, 58.8]	46.2 [41.2, 50.0]
	80%	38	55.0 [45.0, 65.0]	56.2 [50.0, 62.5]	50.0 [43.8, 56.2]	52.5 [50.0, 56.2]
	90%	43	55.0 [45.0, 65.0]	55.0 [48.8, 62.5]	48.8 [43.8, 52.5]	48.8 [45.0, 52.5]
	100%	48	52.5 [43.8, 61.3]	57.5 [51.2, 63.7]	45.0 [40.0, 48.8]	52.5 [50.0, 56.2]
Qwen3-4B	10%	4	41.2 [32.5, 50.0]	51.2 [45.0, 57.5]	42.5 [33.8, 50.0]	51.2 [47.5, 55.0]
	20%	7	52.5 [42.5, 62.5]	52.5 [46.2, 58.8]	53.8 [45.0, 62.5]	50.0 [46.2, 53.8]
	30%	11	58.8 [48.8, 68.8]	51.2 [46.2, 56.2]	63.7 [56.2, 71.2]	46.2 [41.2, 50.0]
	40%	14	46.2 [36.2, 56.2]	51.2 [45.0, 57.5]	55.0 [46.2, 63.7]	46.2 [41.2, 50.0]
	50%	18	43.8 [35.0, 52.5]	55.0 [48.8, 62.5]	45.0 [36.3, 53.8]	52.5 [47.5, 57.5]
	60%	22	51.2 [41.2, 61.3]	50.0 [43.8, 56.2]	55.0 [46.2, 63.7]	50.0 [43.8, 56.2]
	70%	25	47.5 [37.5, 57.5]	53.8 [47.5, 60.0]	50.0 [42.5, 57.5]	52.5 [50.0, 56.2]
	80%	29	48.8 [40.0, 58.7]	52.5 [46.2, 58.8]	51.2 [43.8, 58.8]	52.5 [50.0, 56.2]
	90%	32	47.5 [38.8, 56.2]	52.5 [46.2, 58.7]	52.5 [50.0, 56.2]	48.8 [46.2, 50.0]
	100%	36	50.0 [41.2, 58.8]	52.5 [46.2, 58.7]	51.2 [47.5, 55.0]	50.0 [50.0, 50.0]
Qwen3-14B	10%	4	57.5 [45.0, 70.0]	52.5 [45.0, 60.0]	48.8 [41.2, 56.2]	51.2 [47.5, 55.0]
	20%	8	55.0 [43.8, 66.3]	53.8 [47.5, 60.0]	55.0 [46.2, 63.7]	51.2 [47.5, 55.0]
	30%	12	56.2 [43.8, 68.8]	55.0 [48.8, 62.5]	55.0 [47.5, 62.5]	51.3 [47.5, 56.2]
	40%	16	53.8 [42.5, 65.0]	52.5 [46.2, 58.7]	45.0 [37.5, 51.2]	50.0 [46.2, 53.8]
	50%	20	55.0 [43.8, 66.2]	52.5 [46.2, 58.7]	50.0 [43.8, 56.2]	46.2 [41.2, 50.0]
	60%	24	53.8 [42.5, 65.0]	53.8 [47.5, 60.0]	48.8 [43.8, 53.8]	50.0 [46.2, 53.8]
	70%	28	58.8 [46.2, 70.0]	52.5 [46.2, 58.8]	52.5 [46.2, 58.8]	50.0 [46.2, 53.8]
	80%	32	55.0 [45.0, 65.0]	52.5 [46.2, 58.8]	50.0 [43.8, 56.2]	51.2 [50.0, 53.8]
	90%	36	57.5 [46.2, 68.8]	52.5 [46.2, 58.8]	55.0 [50.0, 61.3]	48.8 [46.2, 50.0]
	100%	40	55.0 [43.8, 66.2]	52.5 [46.2, 58.7]	51.2 [45.0, 57.5]	50.0 [50.0, 50.0]

Table 5: Claude prompt-level consistency across all intervention depths. Tie-adjusted AIMES consistency (%) under Claude-Opus-4.8 with pointwise 95% prompt-bootstrap confidence intervals in brackets ($n = 40$ matched prompts per cell). D_x measures Care/Fairness retention and D_y additional Sanctity suppression.

D.2 DETAILED RESPONSE-QUALITY RESULTS

Objective	AIMES > Fixed on Q	AIMES > Prompt on Q	AIMES > Prompt on Relevance
+ Care + Fairness	26/50	29/50	41/50
+ Loyalty + Authority	24/50	22/50	30/50
+ Care + Fairness – Sanctity	26/50	30/50	35/50
Overall	76/150	81/150	106/150

Table 7: **Blind response-quality comparison.** Number of model–depth settings in which AIMES attains a higher mean score than each comparator. Overall quality is defined as $Q = (C + F + R)/3$, where C , F , and R denote coherence, fluency, and relevance. Counts are descriptive rather than significance tests.

We evaluate whether the multi-value control gains of AIMES are associated with changes in generation quality using a blind GPT-5.6-SOL judge. Each response is scored for coherence (C), fluency (F), and relevance (R) on a four-point scale, with overall quality defined as $Q = (C + F + R)/3$. Table 7 summarizes the number of model–depth–objective settings in which AIMES attains a higher mean score than each comparator. Across the 150 settings, AIMES exceeds Fixed Multi-Value Steering on Q in 76 settings and Prompt Steering in 81, indicating broadly comparable overall response quality rather than a consistent quality advantage. For relevance, AIMES exceeds Prompt Steering in 106/150 settings (70.7%). Taken together, these results do not indicate a systematic loss in response quality under adaptive steering.

D.3 LAYER-WISE WITHIN-RESPONSE STEERING VARIATION

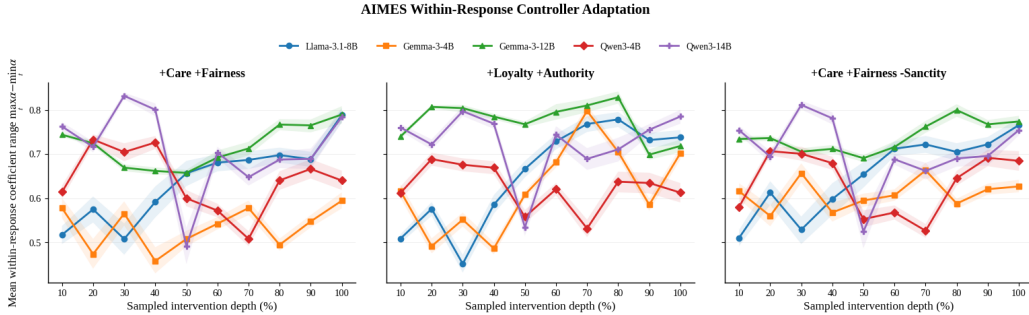


Figure 11: **Variation in AIMES steering strength across intervention depths.** For each model and objective, we measure the within-response range of the target-value steering coefficients, $R_{\alpha,k} = \max_t \alpha_{k,t} - \min_t \alpha_{k,t}$. Each point shows the mean coefficient range across target-disjoint responses at the corresponding model-specific intervention layer, and shaded regions denote 95% prompt-bootstrap confidence intervals. Larger values indicate greater changes in steering strength during generation. Results are reported separately at each of the ten sampled depths; no averaging is performed across intervention layers.

A defining feature of AIMES is that its steering coefficients are updated during generation using the online value readout, rather than remaining fixed for the entire response. We therefore examine how much these coefficients vary within a response across the full intervention-depth grid.

Metric. For each controlled value V_k , we measure the within-response coefficient range

$$R_{\alpha,k} = \max_t \alpha_{k,t} - \min_t \alpha_{k,t}, \quad (26)$$

where $\alpha_{k,t}$ is the steering coefficient applied to value V_k at decoding step t . For objectives involving multiple target values, we first average $R_{\alpha,k}$ across the controlled values within each response and

then average across target-disjoint prompts. Thus, larger values indicate greater changes in steering strength over the course of generation, while smaller values indicate more nearly constant steering. The analysis is performed independently at each sampled intervention layer; no averaging is performed across layers.

Depth-wise analysis. Figure 11 reports the resulting coefficient variation across all ten sampled intervention depths (10%–100%) for the three steering objectives. Across all five models, the coefficient range remains clearly nonzero throughout the network, showing that AIMES continues to adjust its steering strength during generation rather than reducing to a fixed-coefficient intervention. The magnitude of this variation depends on the model, objective, and intervention depth, as expected from a controller whose updates are driven by the model-specific online value state.

The same qualitative behavior is observed across both two-value objectives, ($\uparrow$ Care $\uparrow$ Fairness) and ($\uparrow$ Loyalty $\uparrow$ Authority), as well as the competing three-value objective ($\uparrow$ Care $\uparrow$ Fairness $\downarrow$ Sanctity). Across the plotted conditions, mean coefficient ranges are typically substantial, with values approximately spanning 0.45–0.84. Importantly, this variation is not confined to a single model family or a narrow depth region: all five models exhibit clear within-response changes across early, middle, and later intervention layers.

The representative 30% depth reported in the main text is therefore not an isolated case. Rather, it is part of a broader depth-wise pattern in which the adaptive coefficients remain responsive throughout generation. At this depth, the coefficient range is consistently nonzero across all models and objectives, while the full analysis here shows that this behavior persists across the complete set of sampled intervention layers.

Overall, the layer-wise results confirm that the adaptive component of AIMES remains active across the network: its steering coefficients change meaningfully within responses across models, objectives, and intervention depths, rather than behaving as approximately fixed steering weights.

D.4 INDEPENDENT-JUDGE-BASED ANALYSIS OF DEPTH-WISE AND GEOGAIN RESULTS

Depth-wise interaction contrasts. Figure 12 reports the Claude-based depth-wise advantage of AIMES using the same interaction contrasts as Figure 3. The Care/Fairness-retention advantage over Fixed Multi-Value Steering, D_x , reproduces the strongest depth-specific pattern observed under GPT-5.6-Sol: at 30% depth, $D_x > 0$ for all five models under both judges. The corresponding comparison against Prompt Steering also exhibits 5/5 positive agreement at 30% depth, although the additional depths identified as unanimous under GPT-5.6-Sol do not all remain unanimous under Claude. For example, at 80% depth, Llama-3.1-8B changes sign under Claude ($D_x = -0.113$). The Sanctity-suppression contrast, D_y , shows greater evaluator sensitivity than the Care/Fairness-retention contrast. Against Fixed Multi-Value Steering, the depth at which the advantage concentrates differs between evaluators: under GPT-5.6-Sol, the strongest consensus occurs near 60% depth, whereas under Claude, the strongest consensus for this comparator occurs near 10% depth (Figure 12 (c)). By contrast, the comparison with Prompt Steering is more consistent across evaluators, remaining broadly positive with four of five models positive at several depths under both judges, though without a fully unanimous column under either. Overall, the full-resolution analysis indicates that the 30% Care/Fairness-retention advantage over Fixed Multi-Value Steering is the most evaluator-robust depth-specific finding in this analysis; the Sanctity-suppression advantage over Fixed Multi-Value Steering is depth-dependent in a way that does not consistently localize to the same network region across evaluators, and we scope our main-text claim about this comparison accordingly (Section 4).

GeoGain Analysis. We next recompute the prompt-level GeoGain analysis using Claude scores over the complete ten-depth grid. Table 8 summarizes the number of model–depth cells in which AIMES attains higher GeoGain than each comparator. Across the three objectives and two comparators, this yields $3 \times 2 \times 50 = 300$ model–depth comparisons.

Relative to Fixed Multi-Value Steering, AIMES attains higher GeoGain in 35/50 settings for ($\uparrow$ Care $\uparrow$ Fairness), 29/50 for ($\uparrow$ Loyalty $\uparrow$ Authority), and 30/50 for ($\uparrow$ Care $\uparrow$ Fairness $\downarrow$ Sanctity). The latter exactly matches the 30/50 positive model–depth count obtained with GPT-5.6-Sol, providing independent-judge support for the model- and depth-dependent GeoGain advantage

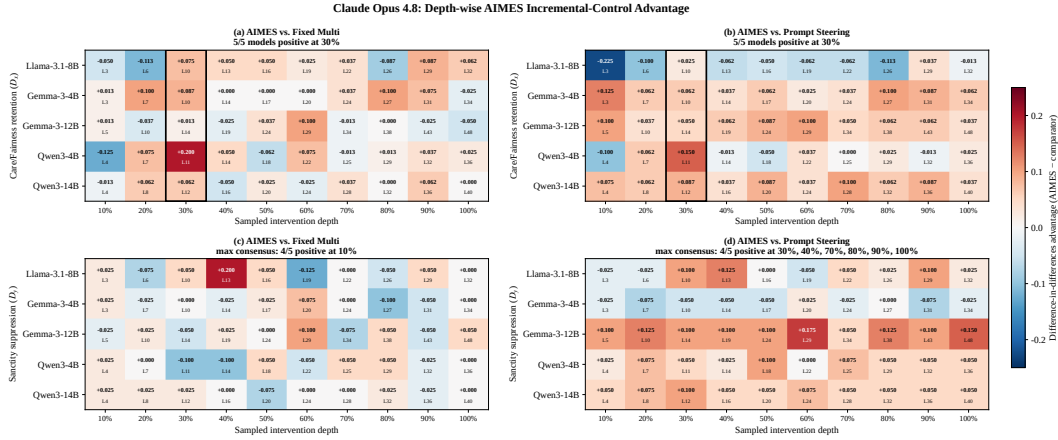


Figure 12: **Claude-Opus-4.8 judge-based analysis of the depth-wise AIMES advantage.** The analysis uses the same ten sampled intervention depths and the same D_x/D_y definitions as Figure 3, but re-scores the identical generations with Claude-Opus-4.8. Positive values indicate an advantage for AIMES; black outlines denote depths at which the advantage is positive for all five models.

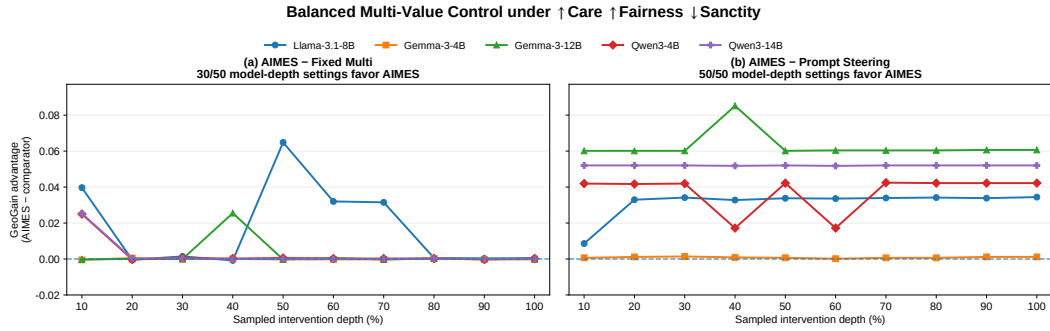


Figure 13: **Claude-Opus-4.8 judge-based analysis of GeoGain for the competing three-value objective.** Prompt-paired GeoGain advantage of AIMES over Fixed Multi-Value Steering (left) and Prompt Steering (right) across all ten sampled intervention depths for ($\uparrow$ Care $\uparrow$ Fairness $\downarrow$ Sanctity). Positive values indicate higher GeoGain metric for AIMES vs. Fixed Multi-Value Steering and Prompt Steering. Under Claude-Opus-4.8, 30/50 model-depth settings favor AIMES over Fixed Multi-Value Steering and 50/50 favor AIMES over Prompt Steering.

over Fixed Multi-Value Steering. Importantly, the 30/50 count understates the asymmetry in effect magnitude: on the three-value objective, cells in which AIMES trails Fixed Multi-Value Steering differ by at most approximately -0.001 in GeoGain, whereas favorable cells reach advantages as large as approximately $+0.065$. Thus, the non-positive cells are predominantly near ties, while several of the positive cells exhibit substantially larger margins. The comparison with Prompt Steering

Objective	AIMES vs. Fixed Multi	AIMES vs. Prompt Steering
($\uparrow$ Care $\uparrow$ Fairness)	35/50	0/50
($\uparrow$ Loyalty $\uparrow$ Authority)	29/50	21/50
($\uparrow$ Care $\uparrow$ Fairness $\downarrow$ Sanctity)	30/50	50/50

Table 8: **Claude-Opus-4.8 GeoGain Analysis.** Number of model–depth cells, out of 50 (five models $\times$ ten depths), in which AIMES attains higher prompt-level GeoGain than each comparator.

is strongly objective-dependent. For ($\uparrow$ Care $\uparrow$ Fairness), AIMES has lower GeoGain in all 50 model–depth settings, whereas for ($\uparrow$ Loyalty $\uparrow$ Authority), it is higher in 21/50 settings. In contrast, for the competing three-value objective ($\uparrow$ Care $\uparrow$ Fairness $\downarrow$ Sanctity), AIMES attains higher GeoGain than Prompt Steering in **50/50** model–depth settings. This strengthens the near-universal advantage observed with GPT-5.6-Sol, where 47/50 settings favored AIMES, and shows that the principal balanced-control result persists under an independent evaluator across the full depth grid. Because GeoGain is positive only when all requested target directions improve simultaneously, the 50/50 AIMES advantage on the three-value objective indicates more favorable joint satisfaction of the competing target directions under Claude, rather than simply larger movement along any individual value dimension.

The model-wise curves further illustrate this reversal. Gemma-3-12B exhibits the largest AIMES advantage over Prompt Steering on the three-value objective, approximately +0.06 to +0.085 across depth, despite AIMES being disadvantaged for this model on the simpler two-value objectives. This suggests that the relative advantage emerges specifically when an additional competing constraint must be satisfied jointly, rather than from uniformly stronger steering on individual target values.

Taken together, the Claude judge-based analysis supports the main GeoGain conclusion while revealing greater evaluator sensitivity in the finer-grained depth-wise decomposition. In particular, the 30% Care/Fairness-retention effect and the balanced-control advantage on the three-value objective persist across evaluators, whereas the precise depth at which Sanctity-suppression advantages emerge is less stable.